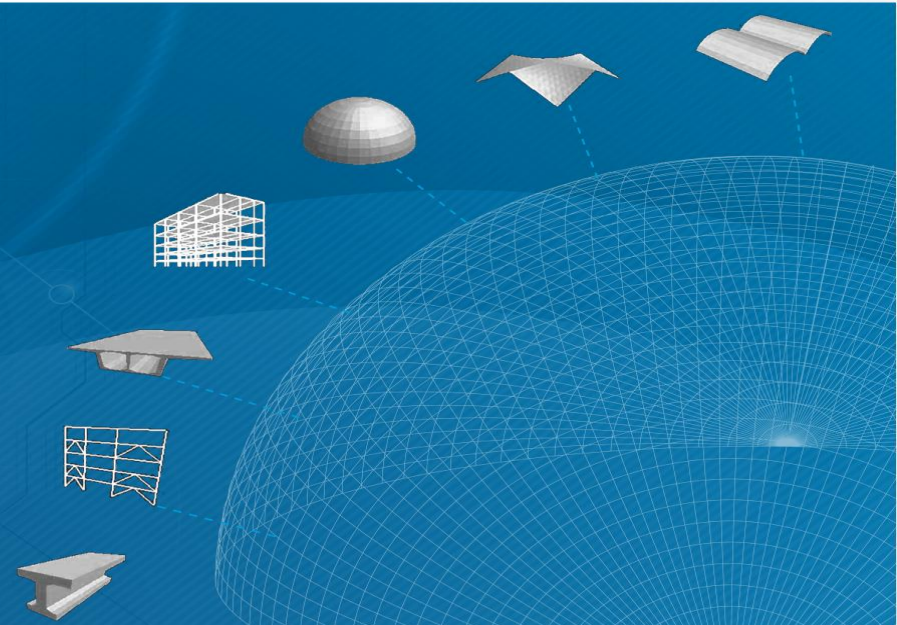
CE - 412 Design of Concrete Structures

Lecture 4:

Cross-sections –

Behavior in Shear and Torsion (Beams)



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Shear Behavior of RC Sections

Frame/ Linear Member Sections

u_z → Axial deformation → Axial strain → Axial stress

u_x → Shear deformation → Shear strain → Shear stress

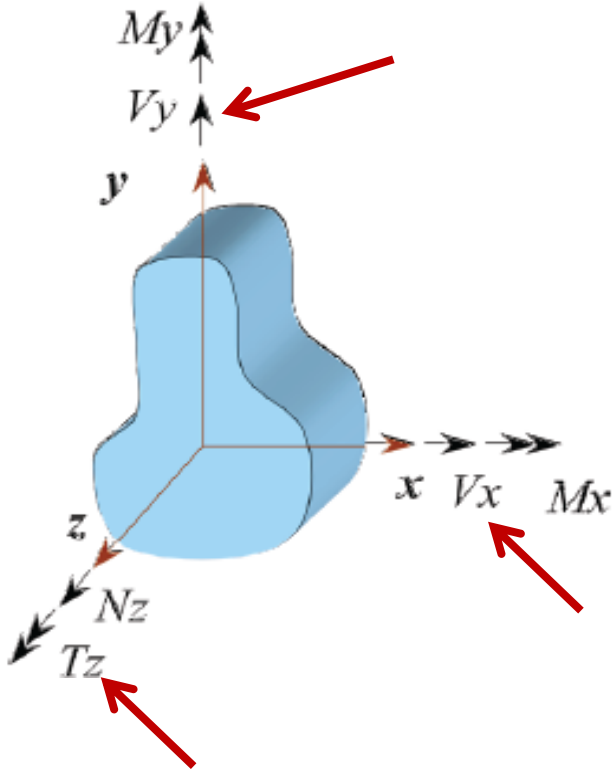
u_y → Shear deformation → Shear strain → Shear stress

r_z → Torsion → Shear strain → Shear stress
(may also produce axial stresses and strains)

r_y → Torsion → Shear strain → Shear stress
(may also produce axial stresses and strains)

r_x → Curvature → Axial strain → Axial stress
(may also produce shear stresses and strains)

w_z → Warping → Axial strain → Axial stress



Shear - Torsion Theories

Major Issues

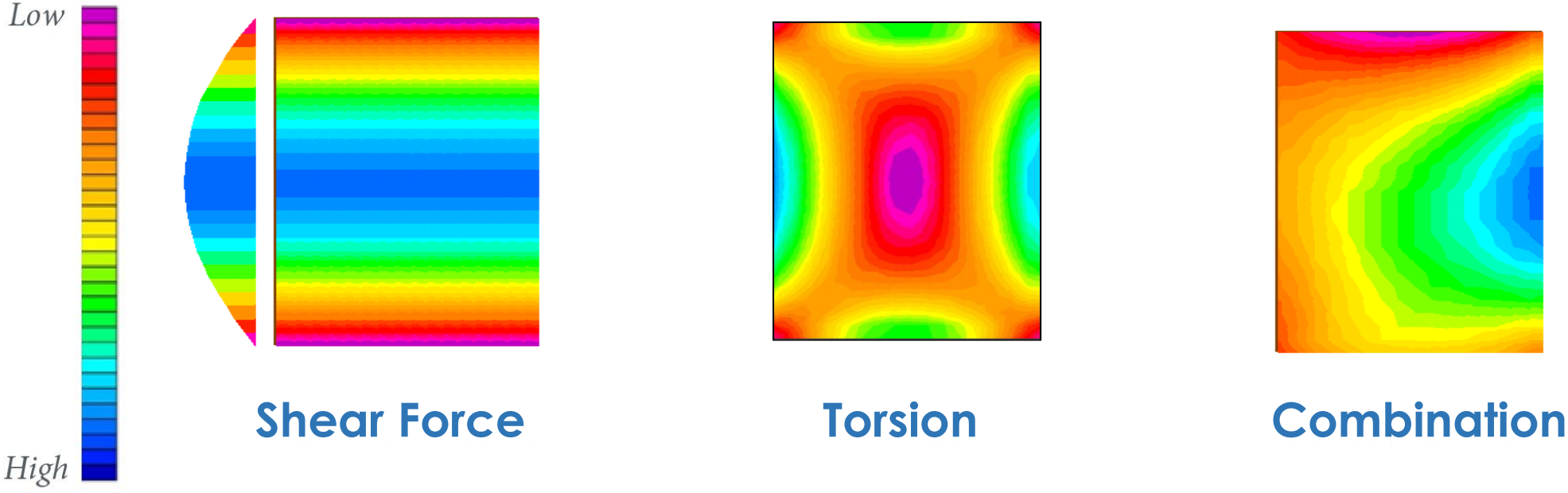
- No unified Theory Yet
- Several Interactions
    - Shear & Torsion, Shear & Moment, Shear & Load, Torsion & Moment
- Effect of Confinement, Flexural Reinforcement, Hoops

Current Approaches

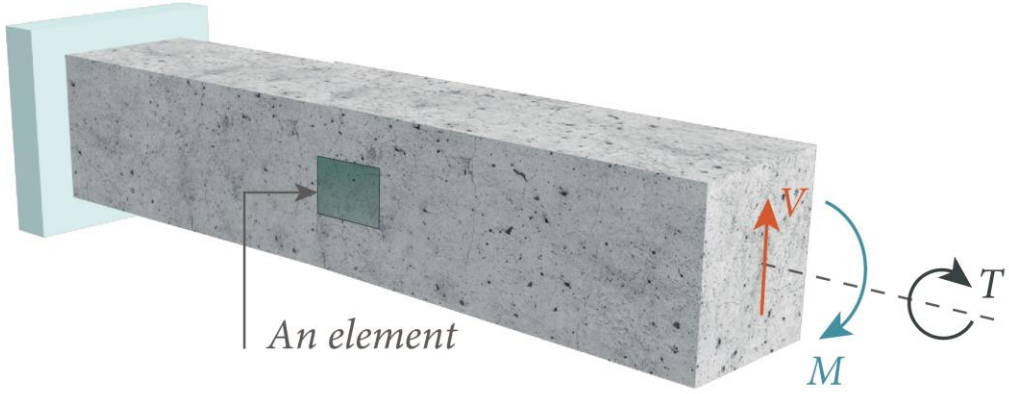
- Solid Mechanics (principle stress, Mohr's circle)
- Truss Mechanism, Strut and Tie, diagonal tension
- Compression Field Theories
- Arch action, dowel action, Fracture Mechanics, aggregate interlock and other considerations

The Origin of Shear Stresses

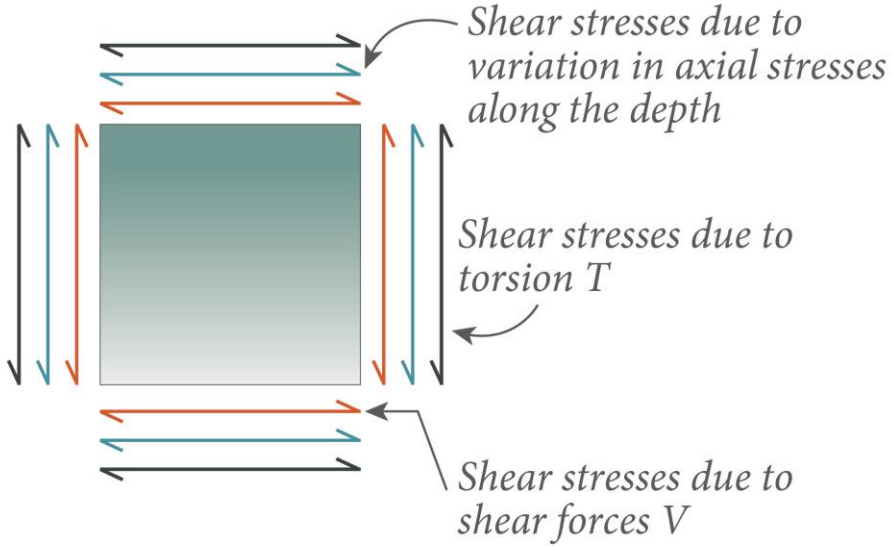
- Shear stress in beam-column sections is produced due to:



The Origin of Shear Stresses



(a) A beam under shear, moment and torsion



(b) Shear stresses in an element due to various actions

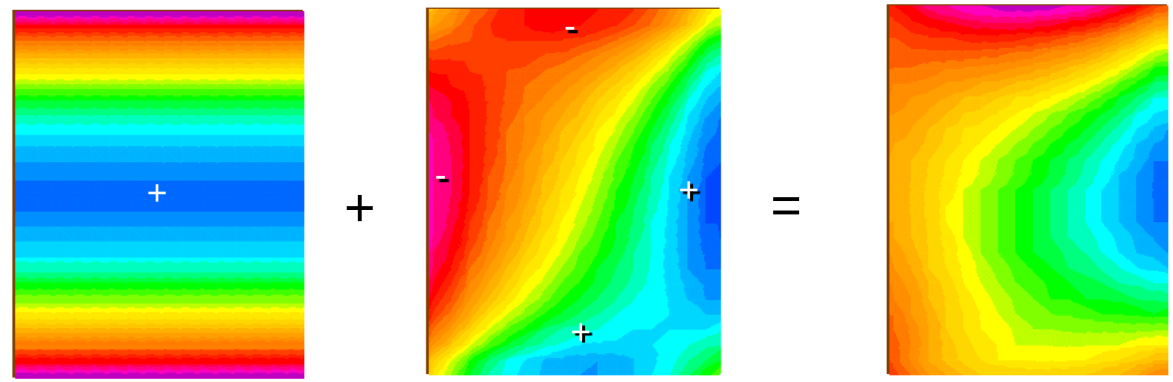
Shear Stress Distribution Due To Shear

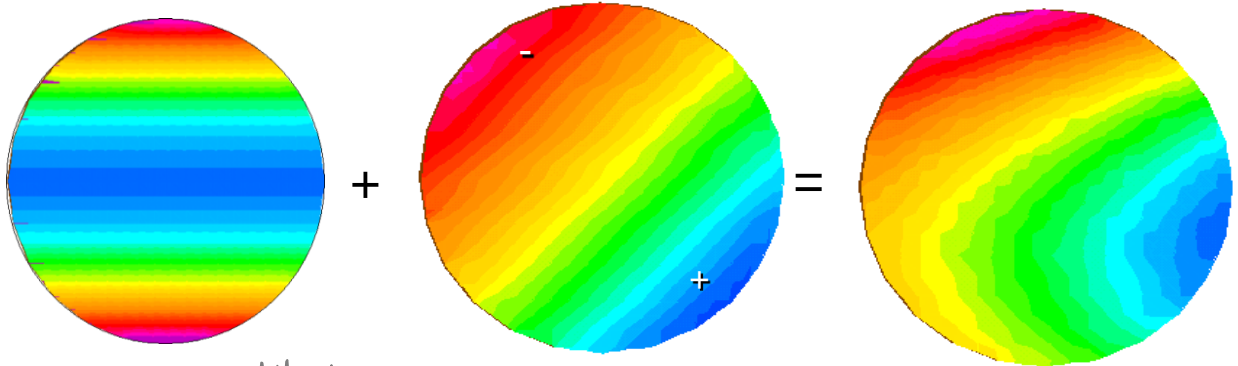


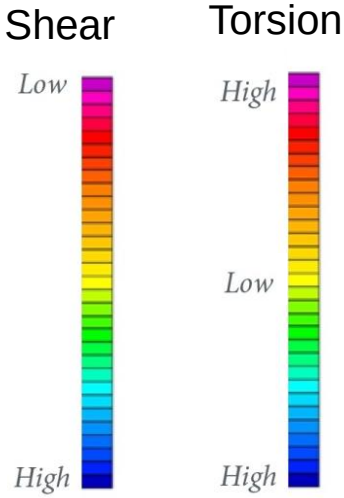
Shear Stress Distribution Due to Torsion



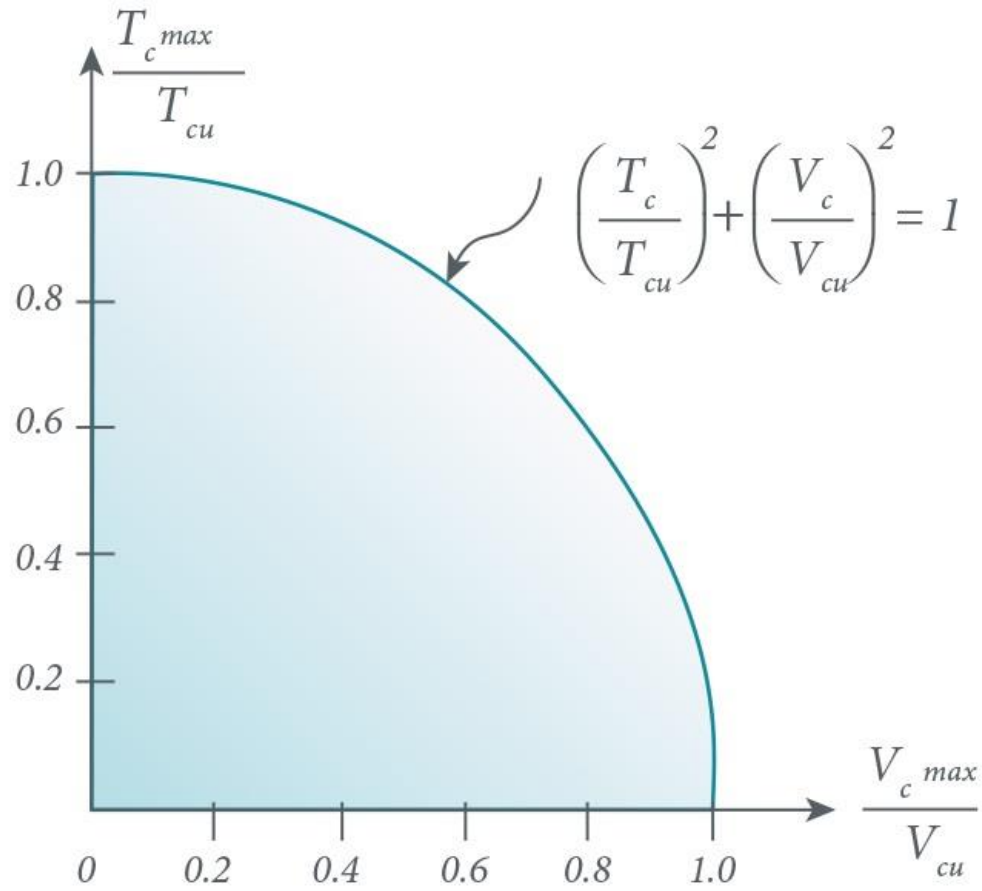
Shear Stress Due to Shear + Torsion



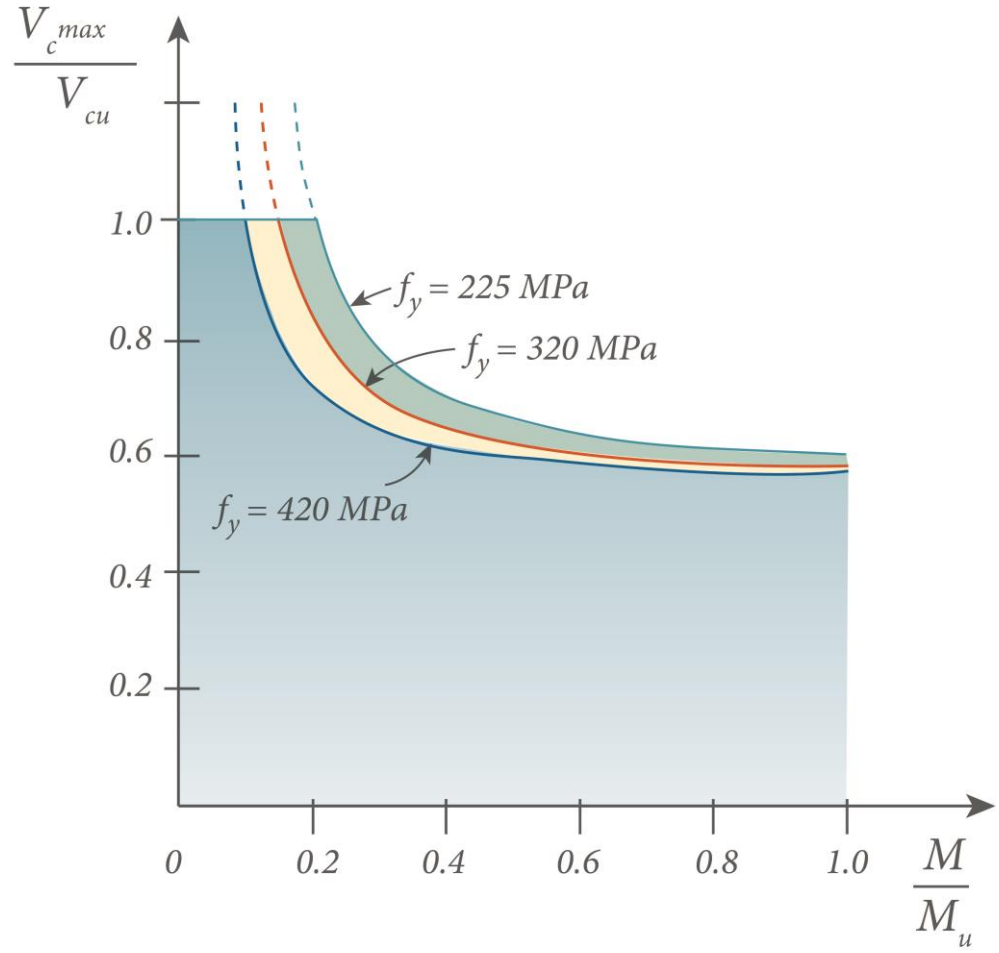




Shear-Torsion Interaction

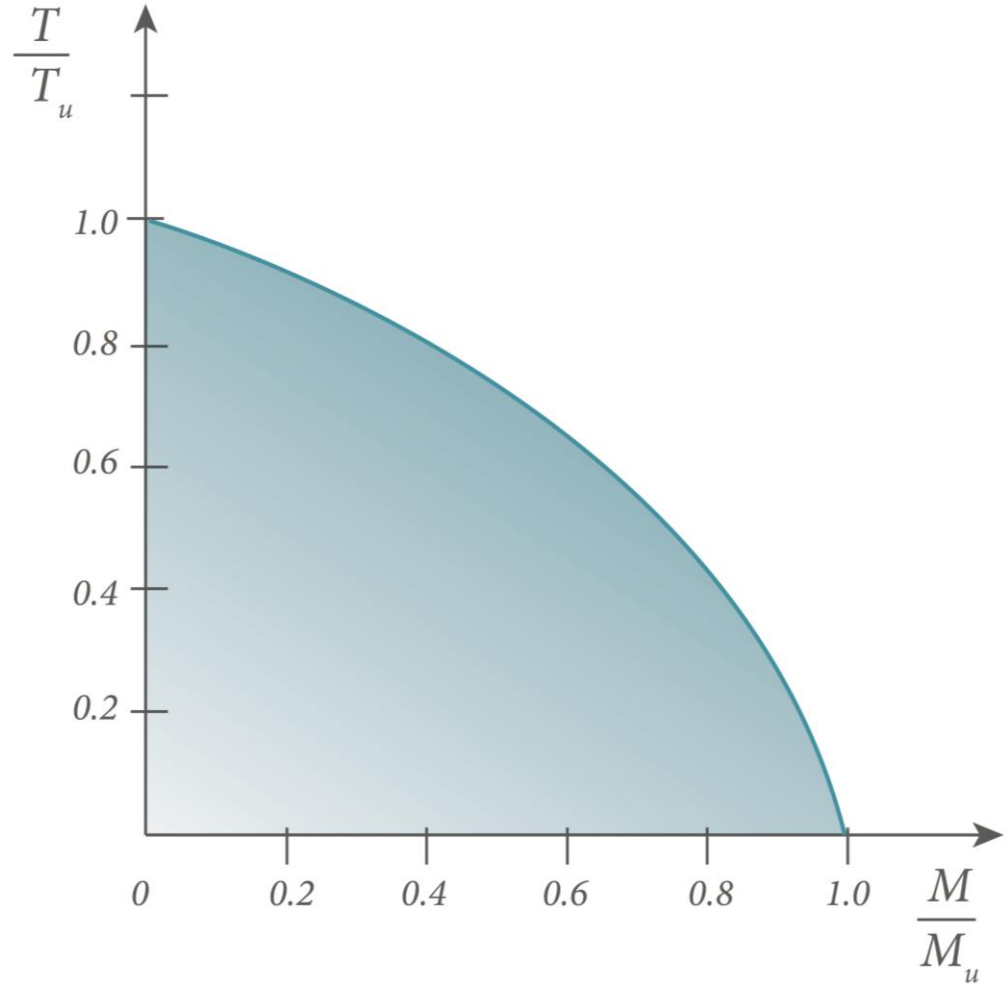


Shear-Flexure Interaction



Torsion-Flexure

Interaction

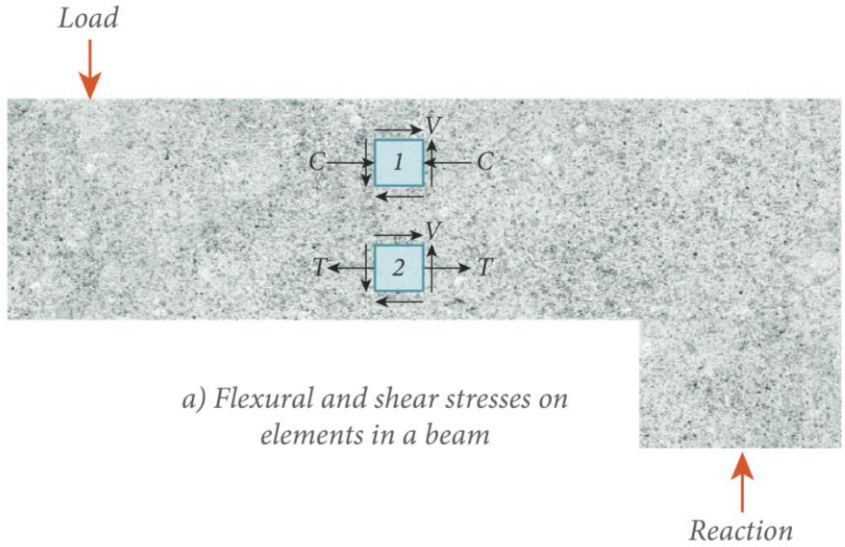


Shear Stress and Concrete Sections

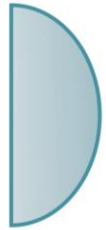
- Shear stress contributes to principal stresses
- In the absence of direct compressive stress:
  - One of the principal stress will be tension
  - Concrete does not take much tensions
  - So concrete can not resist much shear stress
  - Hence concrete alone can not resist much shear force or torsion

- Shear and torsion capacity of concrete section is therefore primarily governed by the tension capacity of concrete.

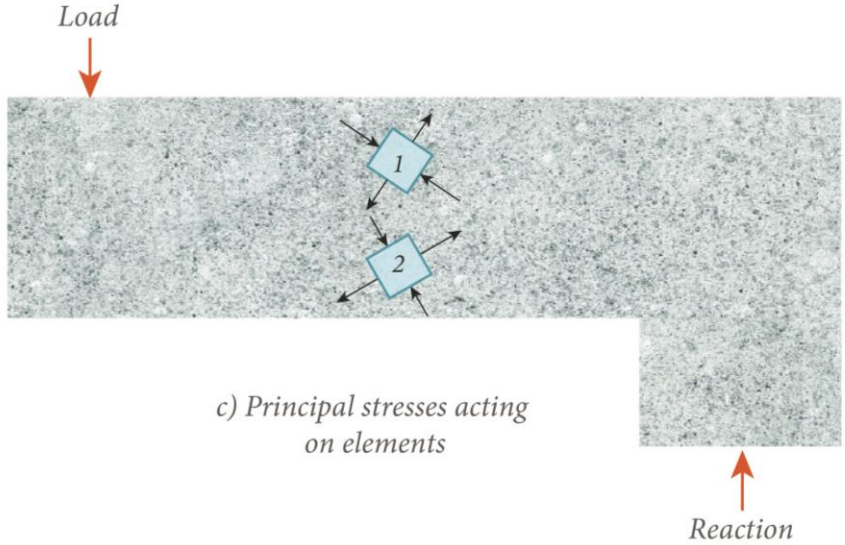
Shear Stress and Concrete Sections



a) Flexural and shear stresses on elements in a beam

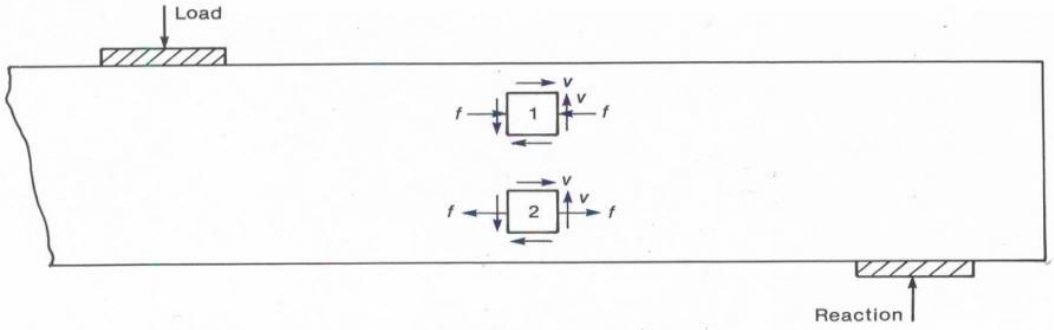


b) Distribution of shear stresses along the depth of section



c) Principal stresses acting on elements

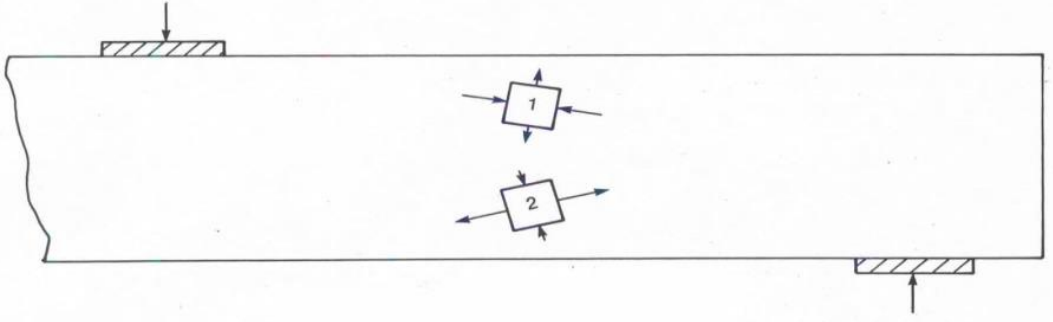
Interaction of Shear Stresses



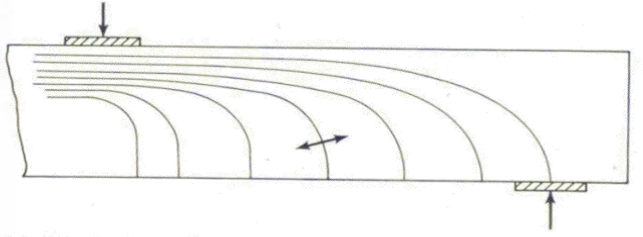
(a) Flexural and shear stresses acting on elements in the shear span.



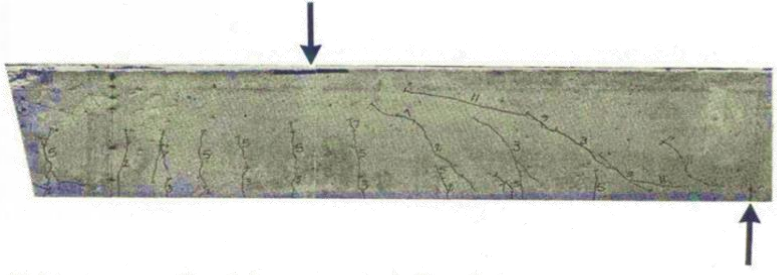
(b) Distribution of shear stresses.



(c) Principal stresses on elements in shear span.



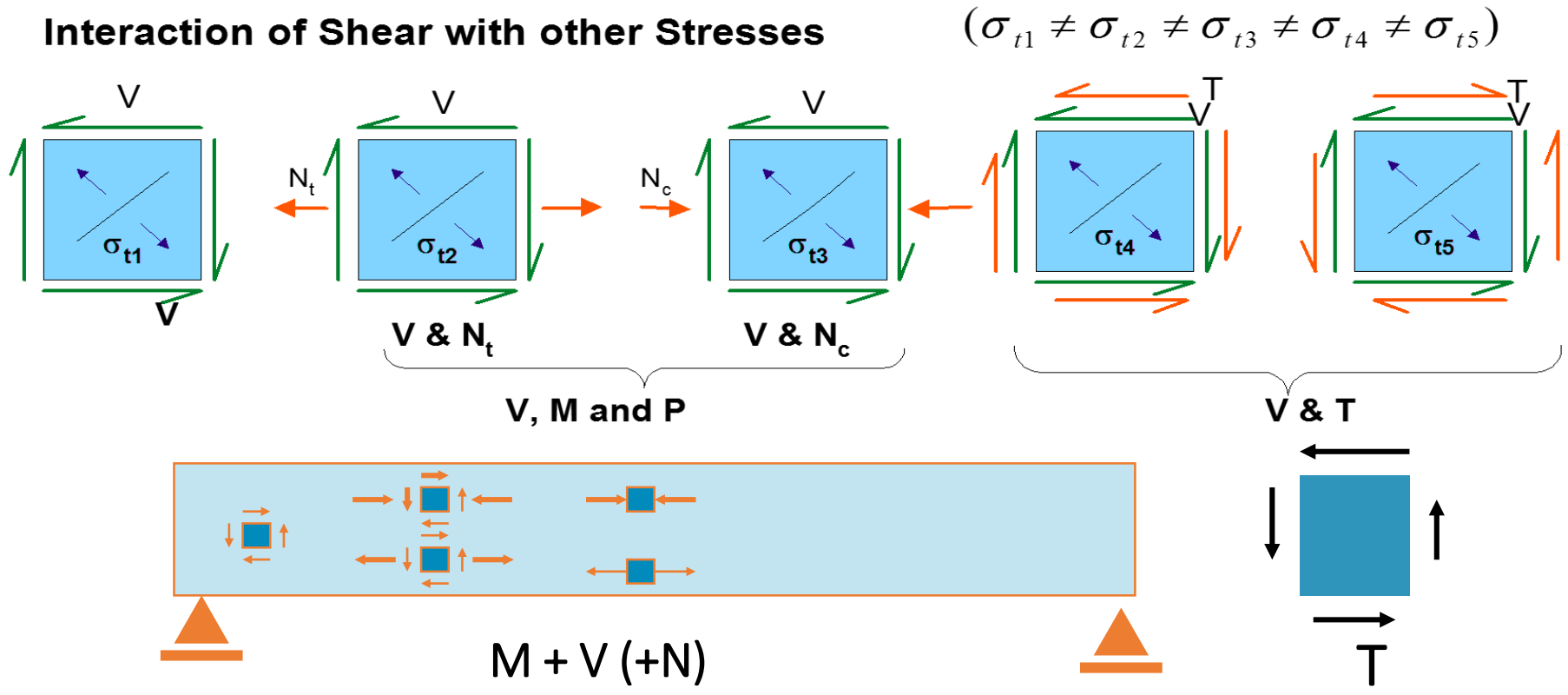
(a) Principal tensile stress trajectories in an uncracked beam.



(b) Photograph of half of a cracked reinforced concrete beam.

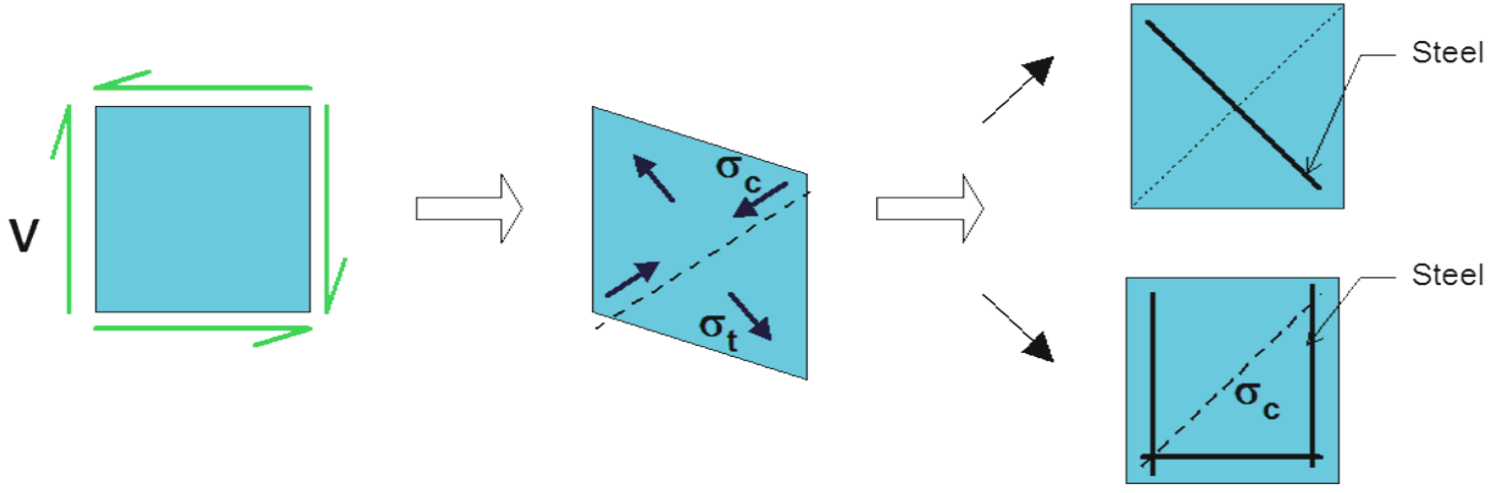
Interaction of Shear Stresses

Consider Interaction of V + T + M + P for computing V_c



Concrete Shear Capacity

Conceptual Design for Pure Shear



Limit V_c on σ_t

$(V_c = 2\sqrt{f_c})$

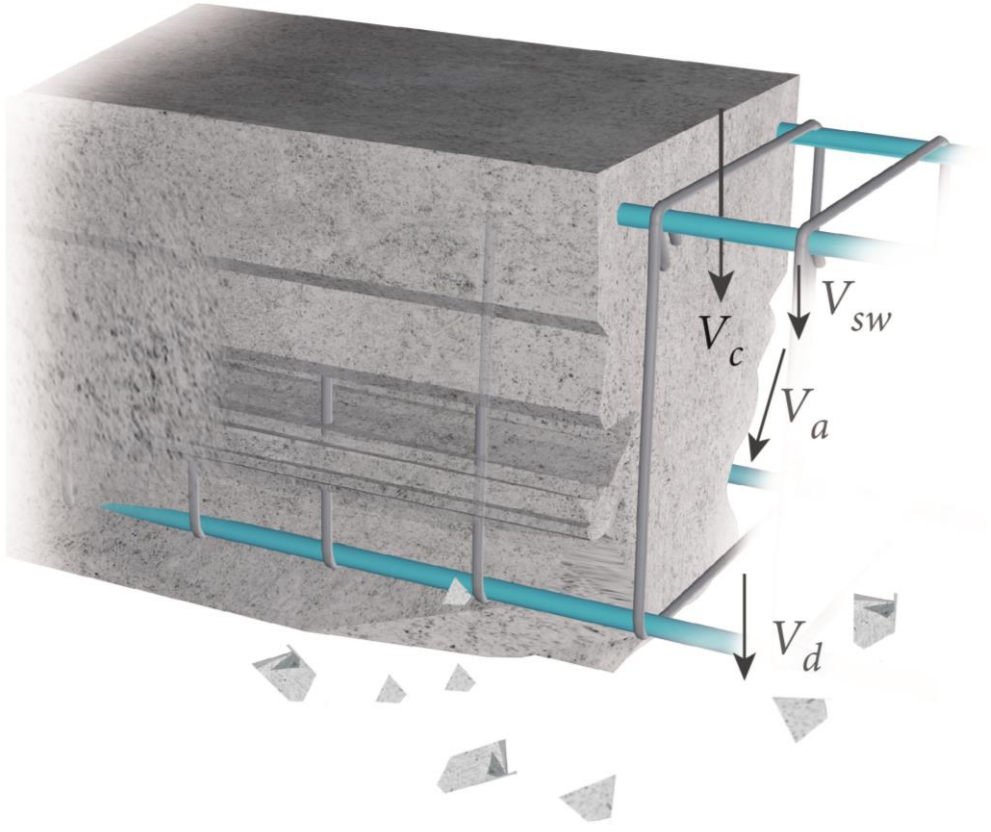
Limit V on σ_c and steel ratio

A_{st} based on V-V_c

Mechanisms of Shear Transfer without Web Reinforcement

- Shear resistance of the un-cracked section above the flexural crack
- Aggregate interlock
- Longitudinal reinforcing to a friction force (dowel action)
- Tied-arch type of behavior that exists in rather deep beams

Shear in Cracked Concrete Beam



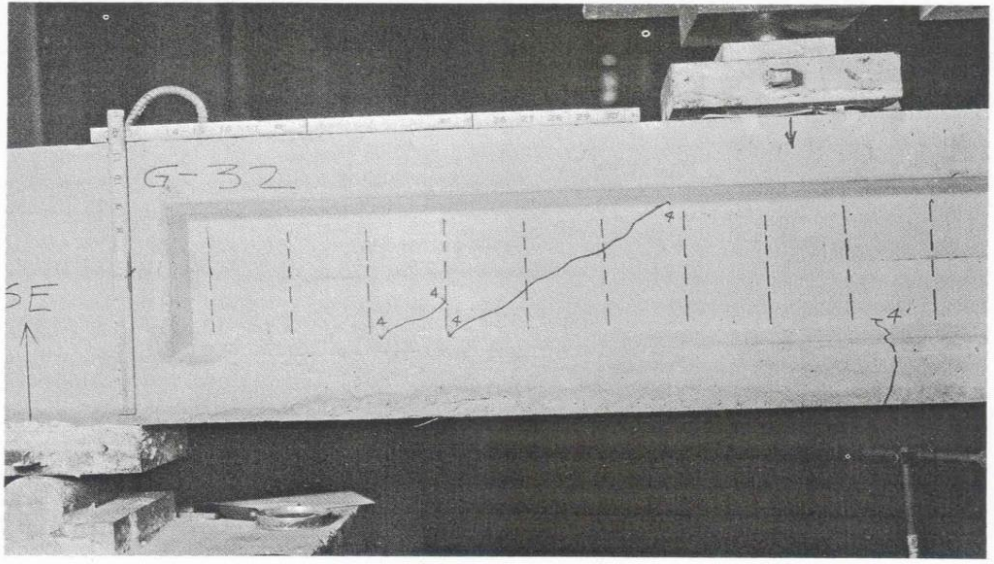
V_c – Intact Concrete

V_a – Aggregate Interlock

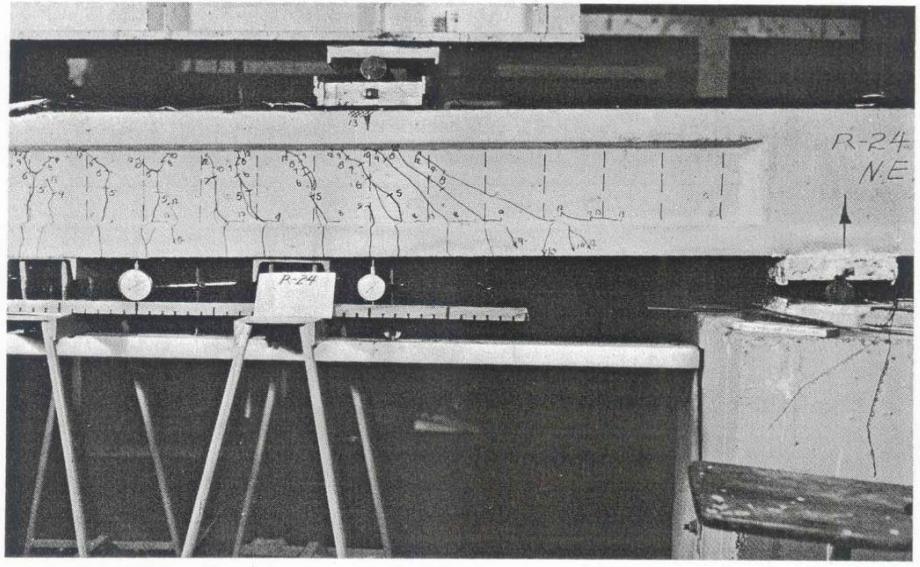
V_{sw} – Steel Reinforcement

V_d – Dowel action

Web and Flexure shear cracks

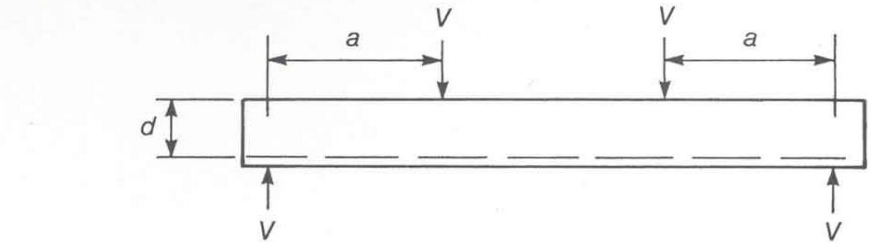


(a) Web-shear crack.

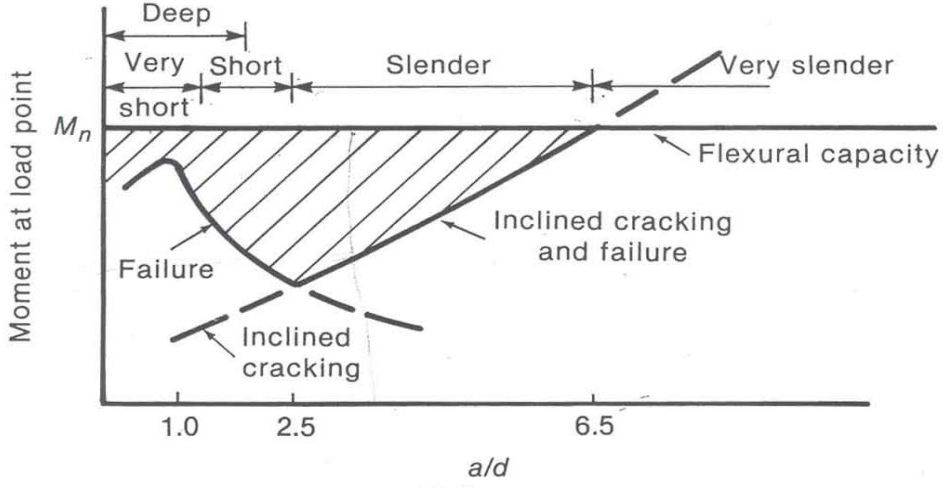


(b) Flexure-shear cracks.

Beams without web reinforcement – Behavior in Shear

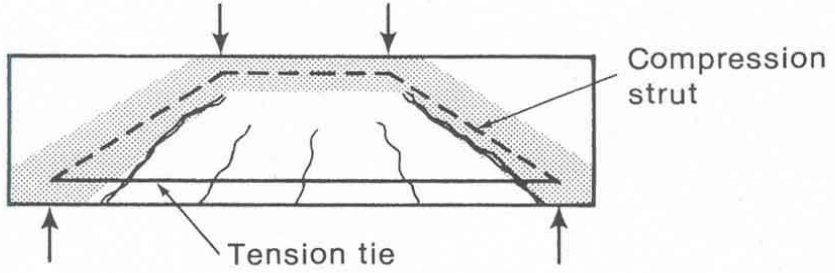


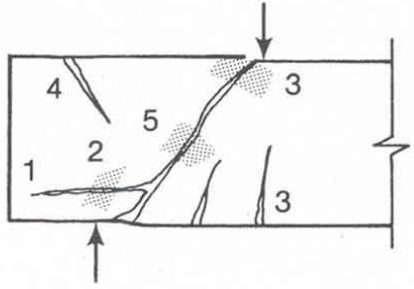
(a) Beam.



(b) Moments at cracking and failure.

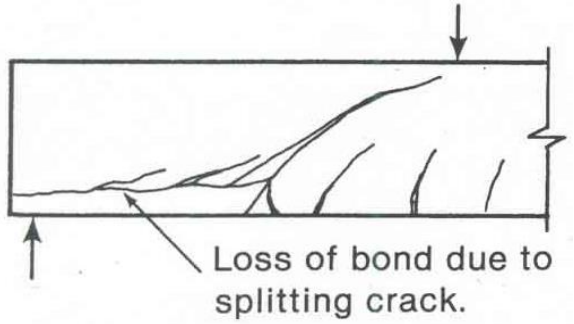
Failures modes of deep beams



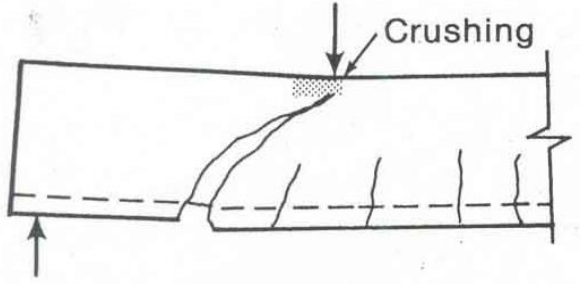


Types of failure:

1 Anchorage failure
2 Bearing failure
3 Flexural failure
4,5 Failure of compression strut

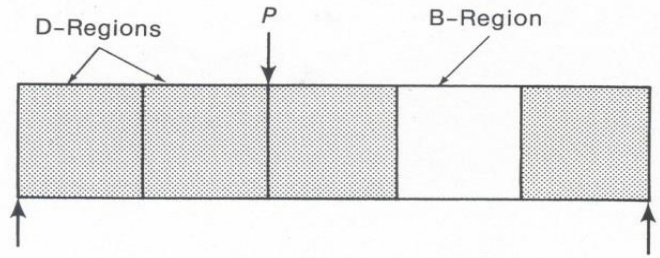


(a) Shear-tension failure.

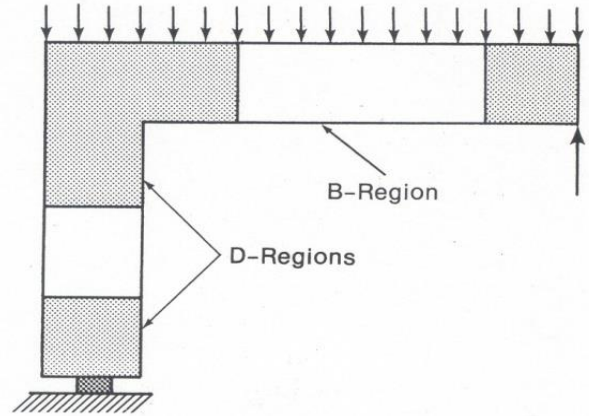


(b) Shear compression failure.

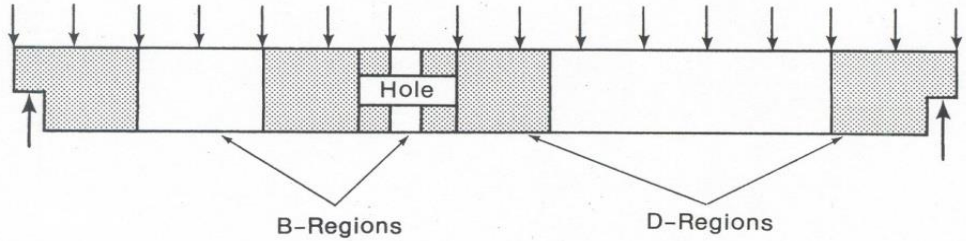
B & D Regions



(a)

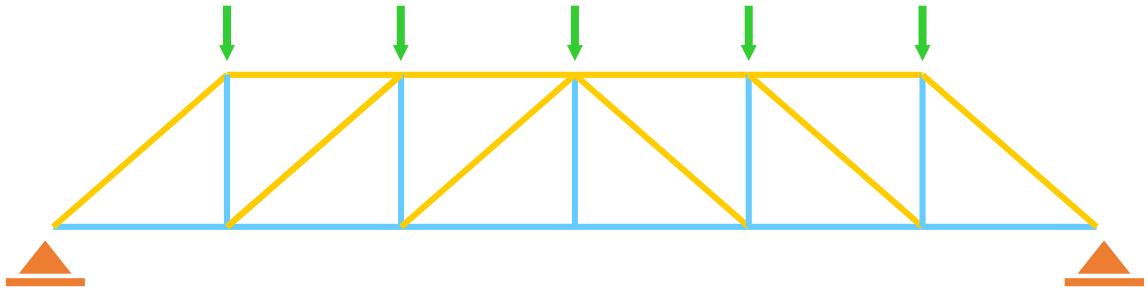


(b)

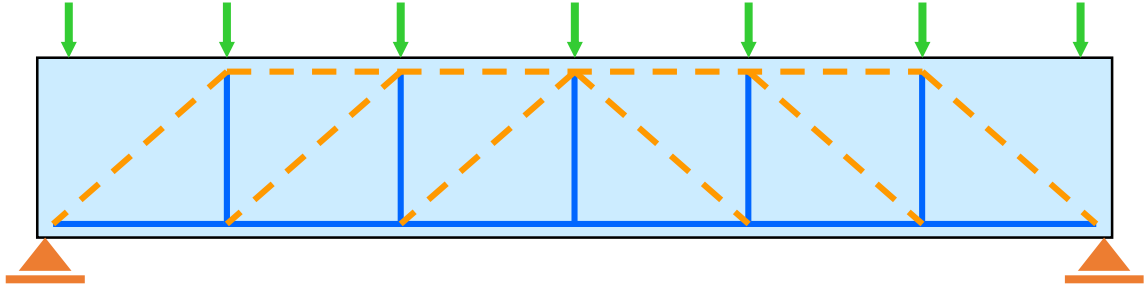


(c)

A Truss and a Beam

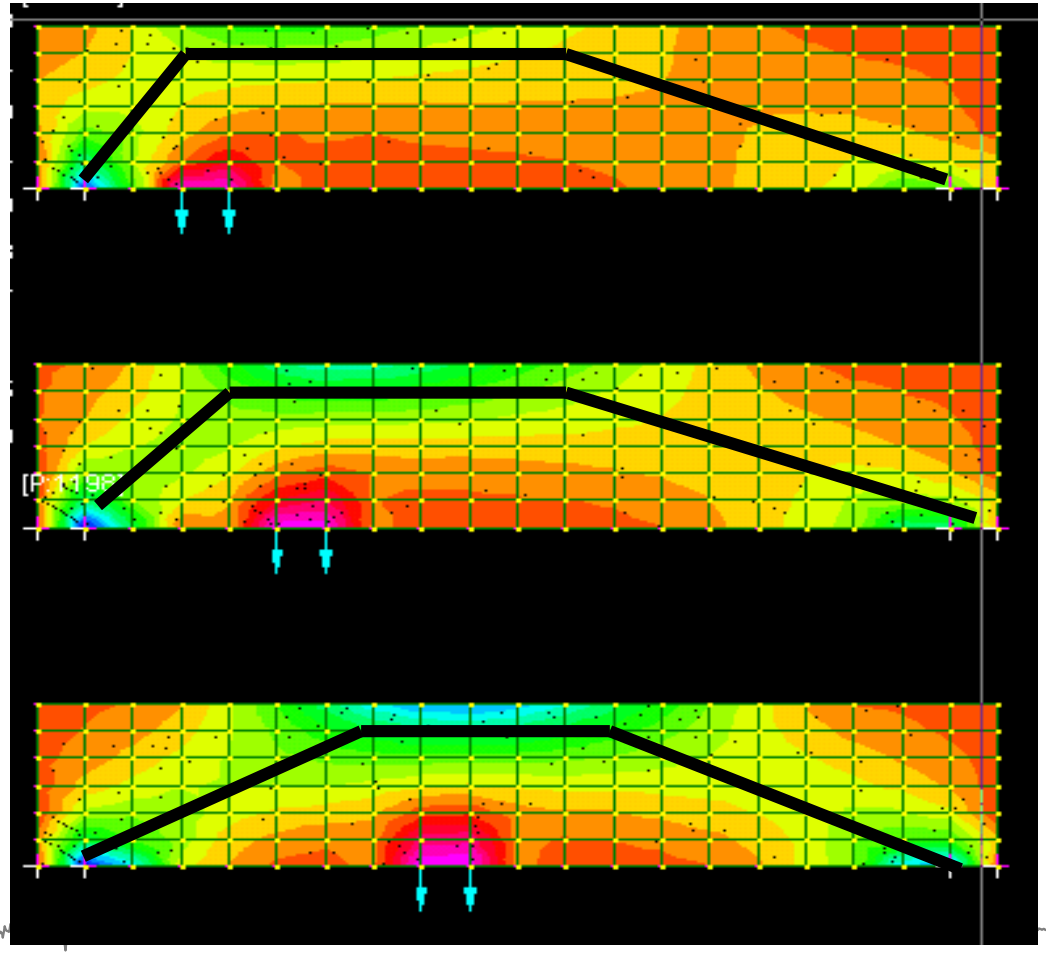


A Real Truss



An RC Beam and “Hidden” Truss

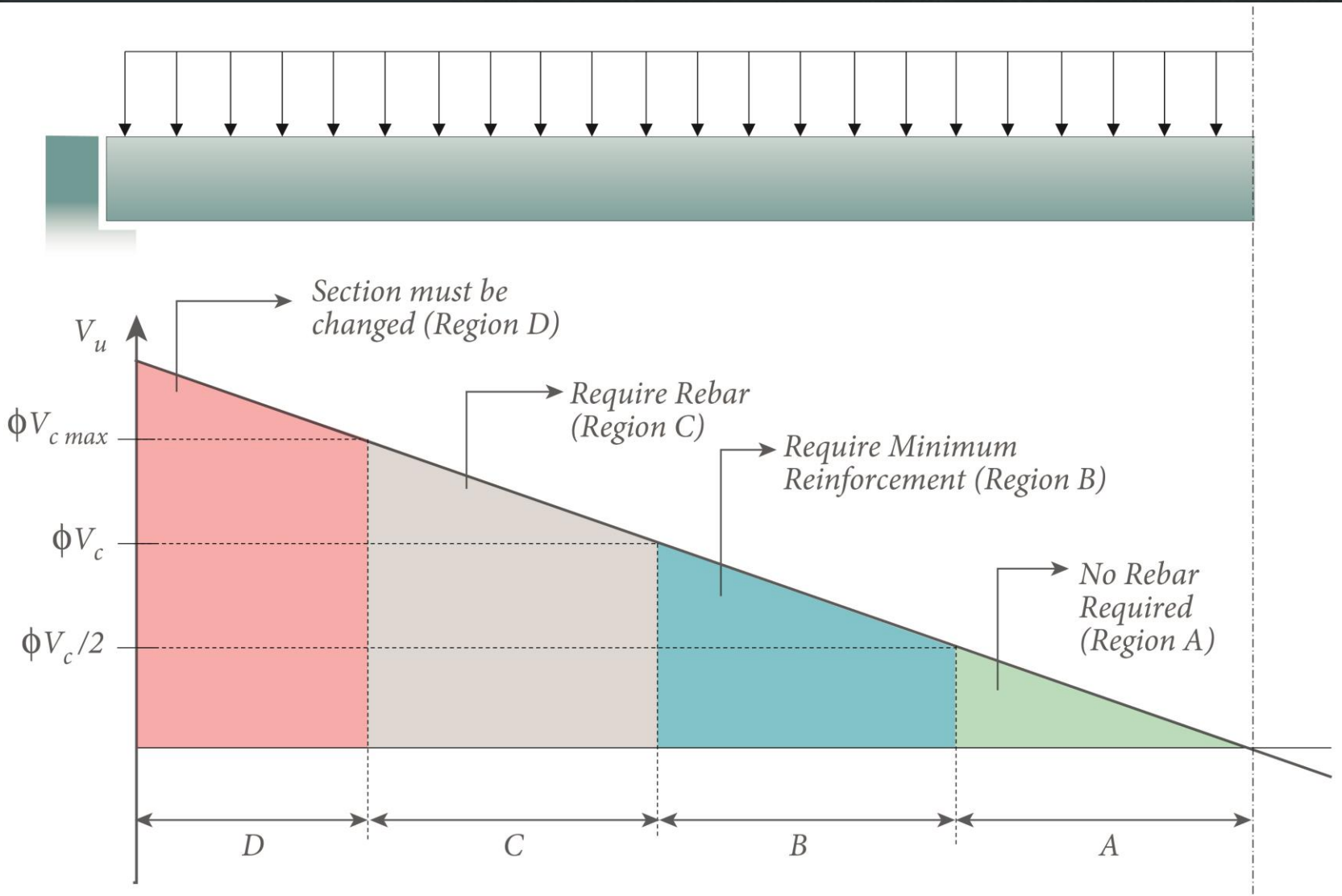
Formation of Truss in Beams



Design for Shear

Design for Shear

- Design practices
- Failure mechanisms
  - Dimensions
  - Geometry
  - Loading
  - Member properties
- Shear strength models
  - Stresses in uncracked beams to explain onset of shear cracking
  - Plastic truss models with shear cracks
  - ACI code design procedure
  - Cracked beam models


Section must be changed (Region D)
V_u
Require Rebar (Region C)
Require Minimum Reinforcement (Region B)
$\phi V_{c\,max}$
ϕV_c
No Rebar Required (Region A)
$\phi V_c/2$
D
C
B
A

Design for Shear

- Compute Cross-sectional properties
- Compute shear capacity of concrete alone
- Determine shear to be resisted by steel
- Compute minimum shear reinforcement
- Compute shear reinforcement separately
- Select rebars and determine bar spacing for shear
- Check spacing limits on rebars



Design for Shear [ACI-318-11]

$$Vn = Vc + Vs$$

- For Axial compression

$$Vc = 2\left(1 + \frac{Nu}{2000\,Ag}\right)\sqrt{fc'}\,bwd$$

- For Axial Tension

$$Vc = 2\left(1 + \frac{Nu}{500\,Ag}\right)\sqrt{fc'}\,bwd$$

- Shear from Stirrups

$$Vs = Av\ fy(\sin\alpha + \cos\alpha)\frac{d}{s}$$

- If $Vu \geq \phi Vn$
  - Stirrups should be provide as

$$\phi V_s \geq V_u - \phi V_c \quad or \quad V_s \geq \frac{V_u}{\phi} - V_c$$

V_c = Shear due to concrete

V_s = Shear due to steel stirrups

V_u = Factored shear load

N_u = Axial Load

A_g = Gross area of steel

f_c' = Concrete strength

d = Effective depth

b_w = Web width

f_y= Yield strength of steel

α = Angle of stirrup

s = Spacing of stirrup

A_v = Area of stirrups

Shear Strength of Concrete (ACI)

- At locations of large moments, uncracked area of beam section reduced

$$V_c < 1.9\sqrt{f_c'}\, b_w\, d$$

- At locations of small moments, large portion of section is available to resist shear

$$V_c \approx 3.5\sqrt{f_c'}\, b_w\, d$$

- Considering effects of longitudinal reinforcements, moments and shear magnitudes, following equation which is less conservative but tedious in calculation, is used

$$V_c = \left(1.9\lambda\sqrt{f_c'} + 2500\rho_w \frac{V_u d}{M_u}\right) b_w d$$

- $V_c = 2\lambda\sqrt{f_c'}\, b_w d$ (Normally used but conservative).

Shear Design (ACI)

- Determine the factored shear force.
- Determine shear force V_c, resisted by concrete.
- Determine the reinforcement steel to carry the balance.
- Check the minimum and maximum shear reinforcement.
- If the required shear reinforcement is more than maximum shear reinforcement, increase the section size.
- Check spacing limits for rebars.

Concrete Shear Strength (V_c) – ACI 318-11

- For members subject to shear and flexure only,

$$V_c = 2\lambda\sqrt{f_c'}\, b_w d$$

- For members subject to axial compression,

$$V_c = 2\left(1 + \frac{N_u}{2000 A_g}\right)\lambda\sqrt{f_c'}\, b_w d$$

- For members subject to significant axial tension,

  - N_u is negative for tension

$$Vc = 2\left(1 + \frac{Nu}{500\, Ag}\right)\sqrt{fc'}\; bw\, d$$

Shear Reinforcement (A_v) – [ACI 318-11]

- If $\phi V_u \leq \phi V_c/2, \quad A_v/s = 0$
- If $\phi V_c/2 < V_u \leq \phi V_{max}$ $\qquad \dfrac{A_v}{s} = \dfrac{V_u - \phi V_c}{\phi f_y \, s \, d}$

where

$$V_{max} = V_c + 8\sqrt{f_c'}\, b_w d$$

- Minimum shear reinforcement

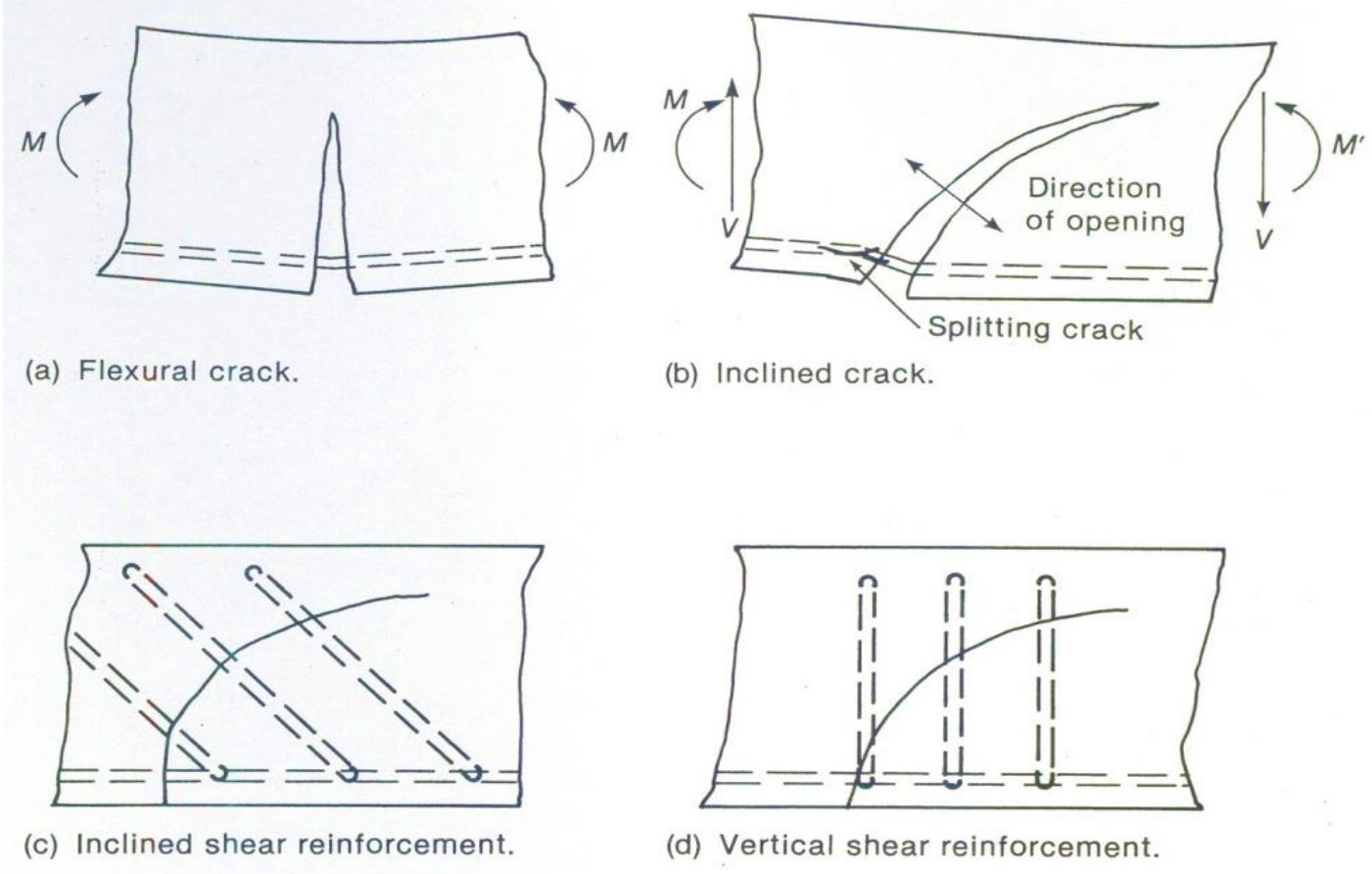
$$A_{v,\,min} = 0.75\sqrt{f_c'}\frac{b_w s}{f_{yt}} \quad < \quad (50 b_w s)/f_{yt}.$$

Shear Reinforcement (A_v)

- V_s is limited by, $8\sqrt{f_c'}\ b_w\ d$ since a beam shear strength cannot be increased indefinitely by adding more and more shear reinforcement.

- The greater the shear transferred by shear reinforcing to the concrete, the greater will be the chance of a combination shear and compression failure.

Shear Reinforcement (A_v)



Rebar Spacing Limit

- Maximum Spacing

Required V_s		Maximum s, in.			
		Nonprestressed beam		Prestressed beam	
		Along length	Across width	Along length	Across width
$\leq 4\sqrt{f'_c}b_w d$	Lesser of:	$d/2$	d	$3h/4$	$3h/2$
		24 in.			
$> 4\sqrt{f'_c}b_w d$	Lesser of:	$d/4$	$d/2$	$3h/8$	$3h/4$
		12 in.			

- Minimum Spacing
  - Approximately 3 or 4 in

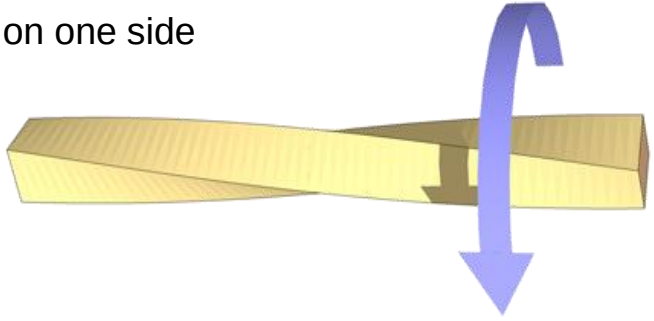
Shear Design in Ductile Frames

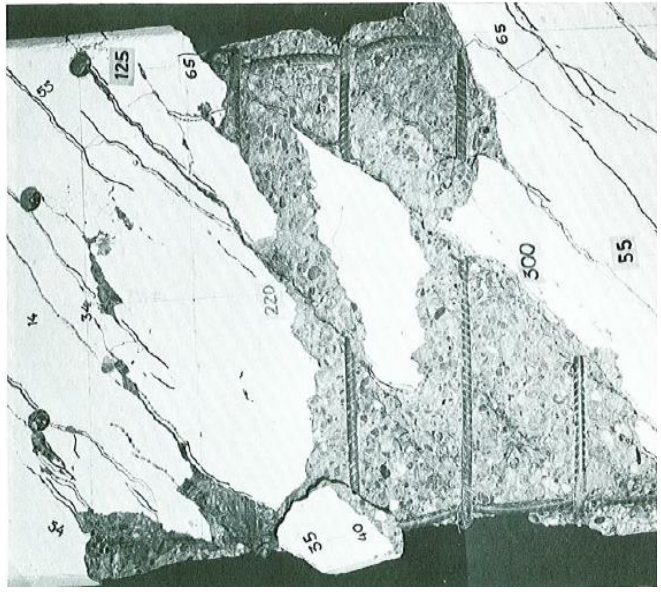
- Design shear force is based on maximum probable moment capacities and gravity shear forces.
- Concrete shear strength is not considered in the hinge region when both of the following conditions occur.
- The earthquake induced shear force represents one-half or more of the maximum required shear strength.

Torsional Behavior of RC Sections

Torsion

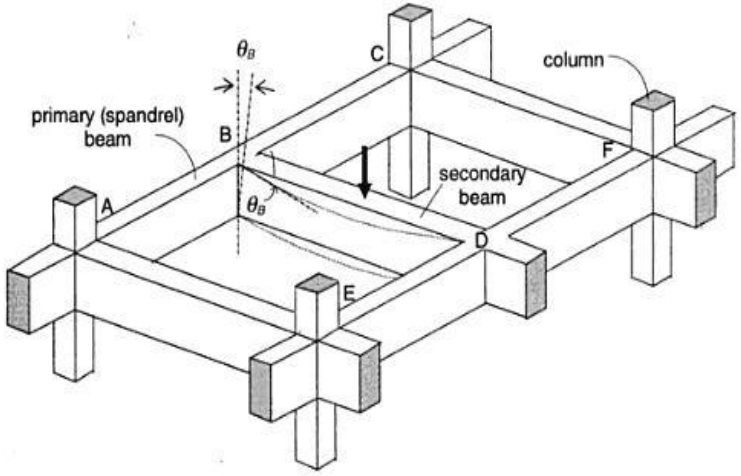
- Twisting of an object due to applied torque / moment
- It is usually not a problem
  - For average designer with large cross sectional areas of beams
- It becomes more pronounced in smaller cross sections
- It causes failure of beams in 45 Degree pattern like shear
  - It adds to shear on one side
  - It subtract from shear on one side





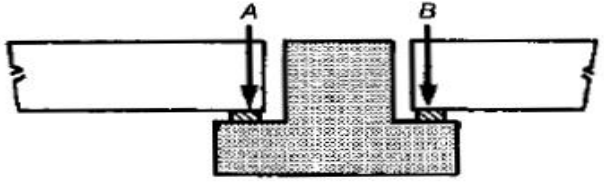
Importance

- Slender beams / girders of a bridge
- Spandrel beams
- Beams in unsymmetrical buildings
- Curved beams
- Off center loads
- Reversal of stresses in earthquake
- Sometimes becomes even more critical than axial, flexural or shear strength

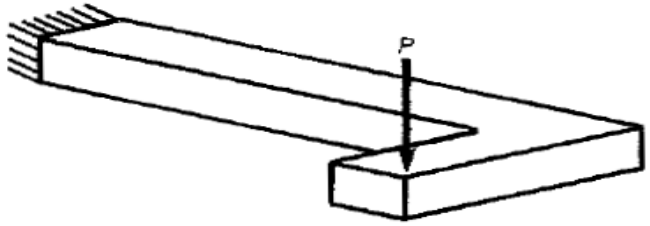




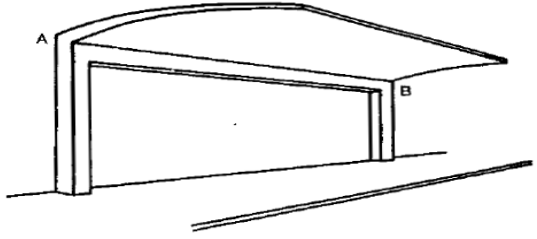
Torsion



Section through a beam supporting precast floor slab



Cantilever with eccentrically applied load



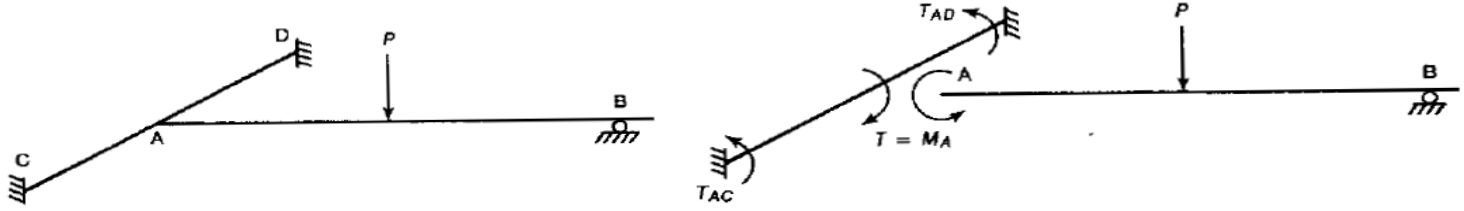
Canopy

Torsion

- Torsion:
  - A moment acting about the longitudinal axis of a member is called a torsional moment. In structure, torsion results from the eccentric loading of beams or from deformations resulting from the continuity of beams or similar members that join at and angle to each other.
  - From the design point of view, torsional moments are classified into equilibrium moments and compatibility torsional moments.
- Equilibrium torsional moment:
  - It is torsional moment against which a structure must resist in order to keep the force equilibrium in the overall structure system. If this torsional moment is neglected in the calculation of the force equilibrium of the structure the stability of the overall structure is compromised.

Torsion

- Compatibility Torsional Moment
  - It is a moment caused by the compatibility between the members meeting at a joint (composing statically indeterminate structure), and influences mainly on the elastic deformation of the structure.



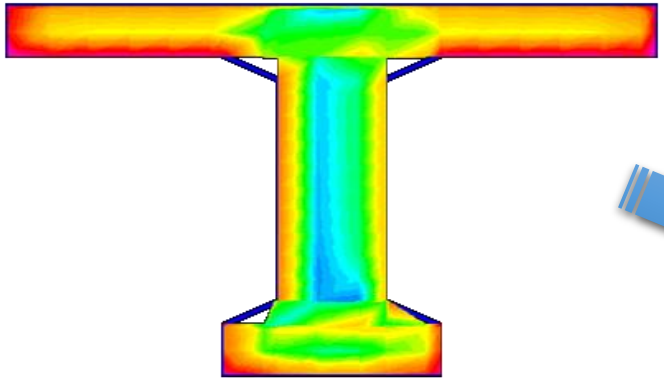
- In general, the torsional rigidity of a concrete member is greatly reduced after plastic deformation by torsion. Hence, the torsional moment action on the member becomes very small when a concrete member of a statically indeterminate structure reaches such a state.

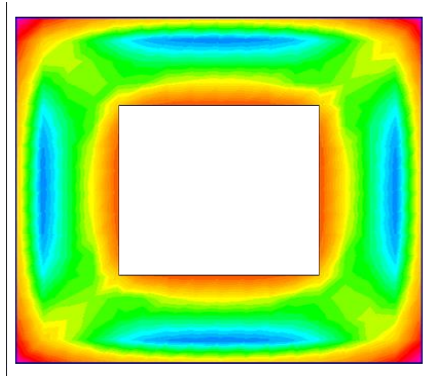
Torsional Reinforcing

- Like shear stresses, provision of torsional reinforcement certainly reduce the chances of abrupt failure.
  - Actually sometimes only stirrups are sufficient
  - For torsion, combination of longitudinal steel with stirrups is a desired combination
- Provision of only U stirrups can reduce chances of shear failure but are not suitable for torsion
  - Ideally a closed stirrup with 135 degree hooks
  - Essentially encompass the longitudinal steel at corners
  - The diameter of longitudinal bar should not less than #3 or 0.042 times stirrup spacing (ACI 11.6.6.2)

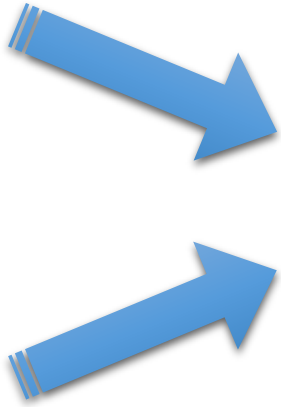
Torsional Stress Distribution

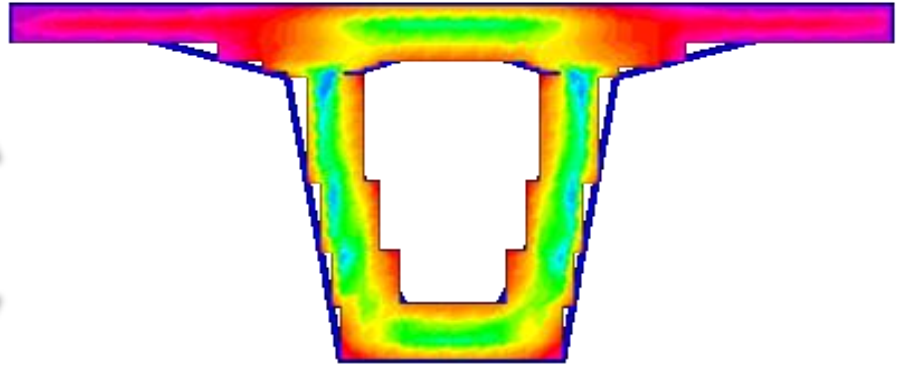
A Flanged Section





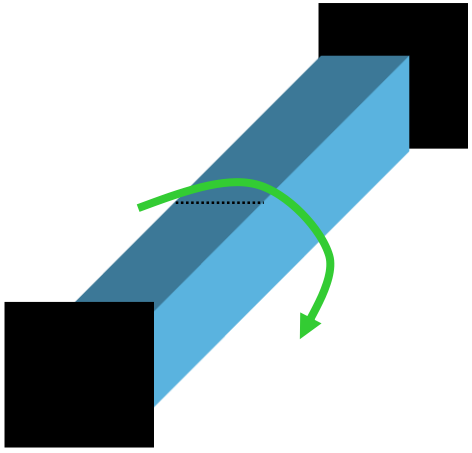
A Box Section

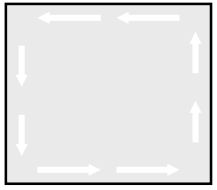




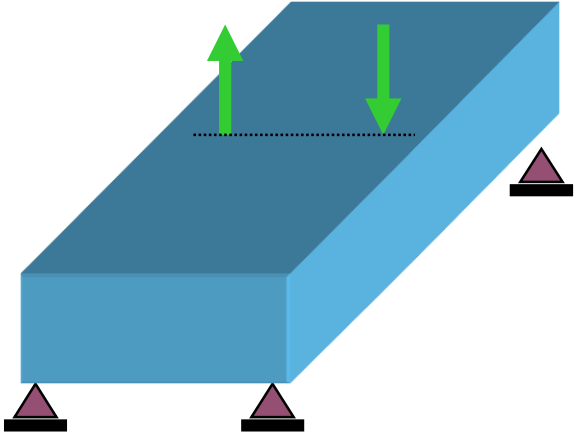
A Flanged - Box Section

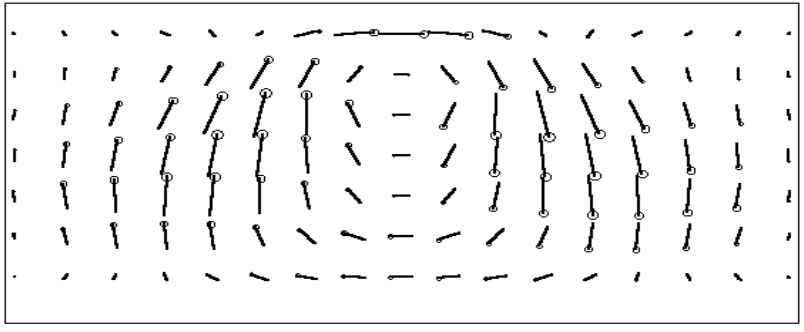
Behavior for Shear and Torsion





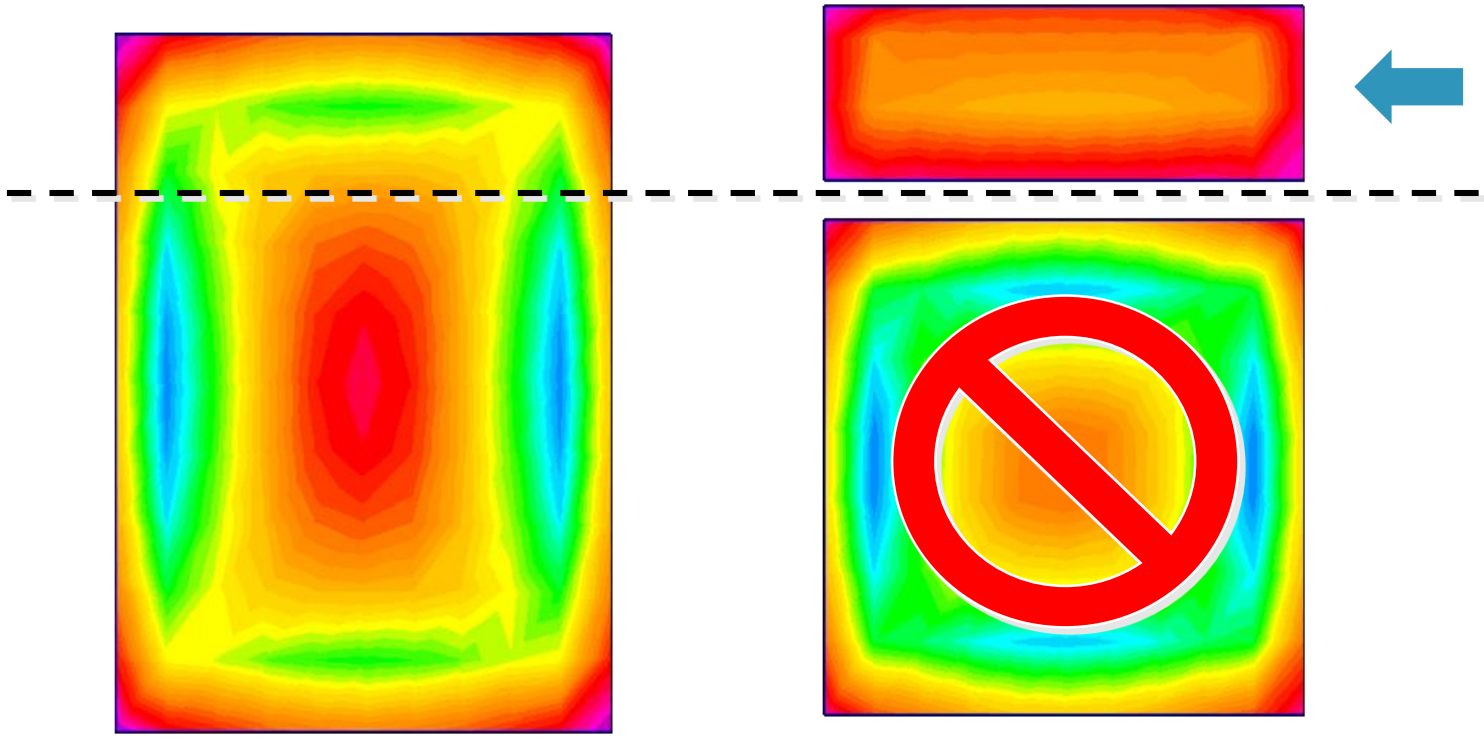
Pure Torsion produces constant “Shear Flow” around the cross-section outer skin.





Shear and Torsion may produce Un-even Shear Flow

Simplified Behavior for Torsion – RC Beam



Before Cracking of Concrete

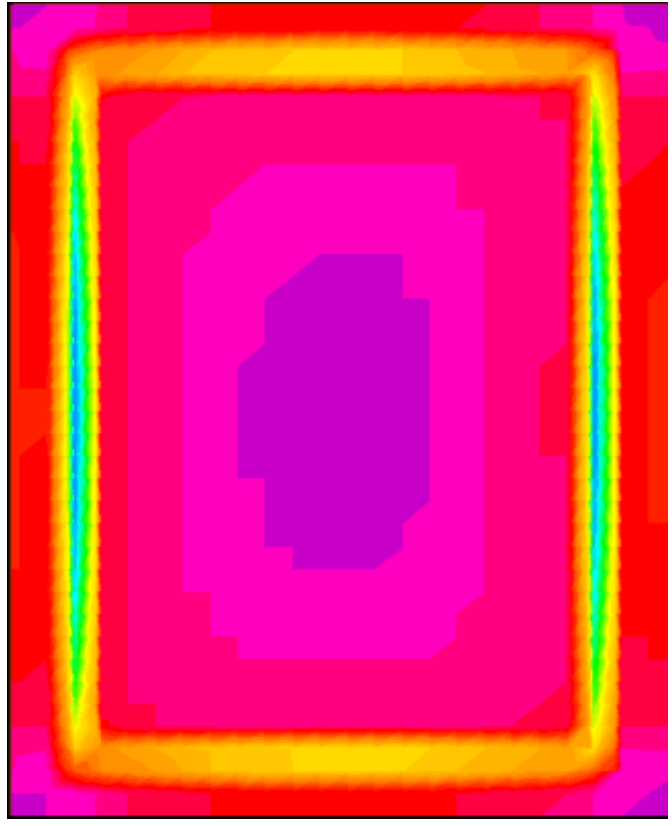
After Cracking of Concrete

Only this portion of concrete section is available for resisting Torsion

Therefore “other” mechanism should be found to resist torsion

Simplified Behavior for Torsion – RC Beam

- The shear flow in a steel tube embedded in a concrete section is concentrated in the steel tube.
- This observation is relevant to concrete sections with hoop reinforcement, where almost the entire torsion is resisted by the reinforcement.
  (green color show high stress)



Torsional behavior before cracking

- The mechanical behavior of a concrete member before torsional crack can be estimated by using elasticity theory, assuming that the gross concrete section is effective.
- The angle of twist per unit length is

$$\theta = \frac{T\,L}{G_c\,J_1}$$

  - Where T = twisting moment,
  - G_c= the shear modulus of concrete
  - J_1= the torsion constant.
  - For rectangular section,

$$J_1 = cb^3\left[\frac{1}{3} - 0.21\frac{b}{c}\left(1 - \frac{b^4}{12c^4}\right)\right]$$

where c and b are the two sides of the rectangle with b<= c.

The maximum shear stress is at the middle of the longer side c and its value

$$\tau_{\max} = \frac{T}{\mu bc^2}$$

Torsional behavior before cracking

- where μ is a dimensionless coefficient which varies with the aspect ratio c/b as

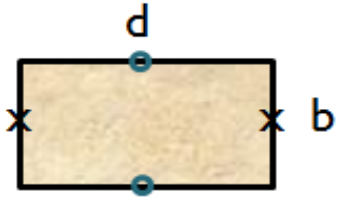
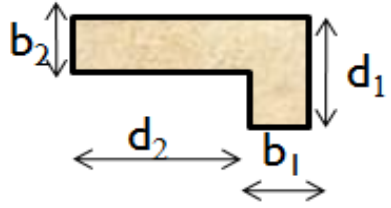
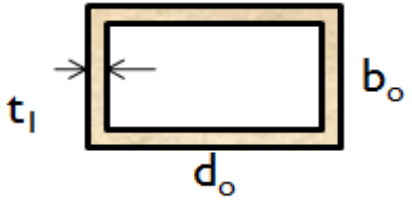
c/b	1.0	1.5	1.75	2.0	2.5	3.0	4.0	6.0	8.0	10.0	∞
μ	0.208	0.231	0.239	0.246	0.258	0.267	0.282	0.299	0.307	0.313	0.333

- Torsional moment (Mt) acting on the member section is given by

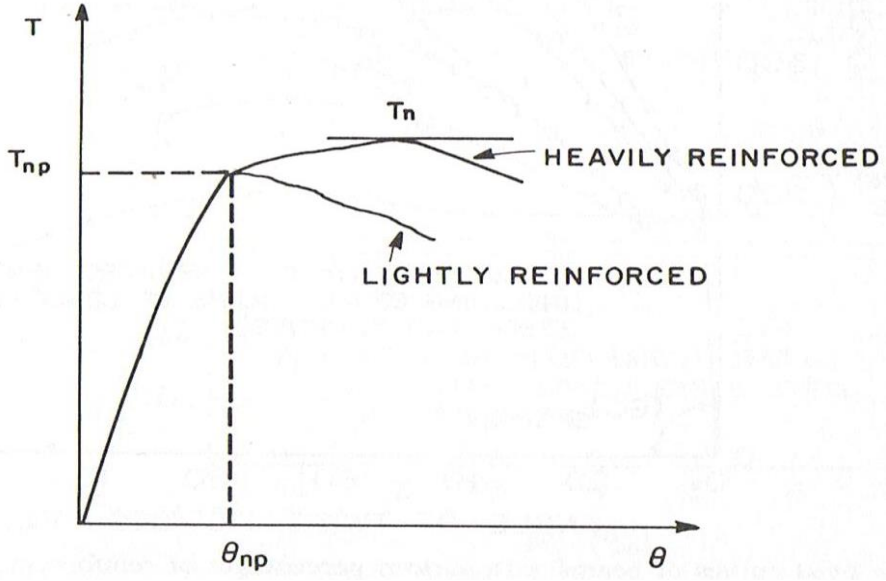
$$M_t = K_t \tau_{\max}$$

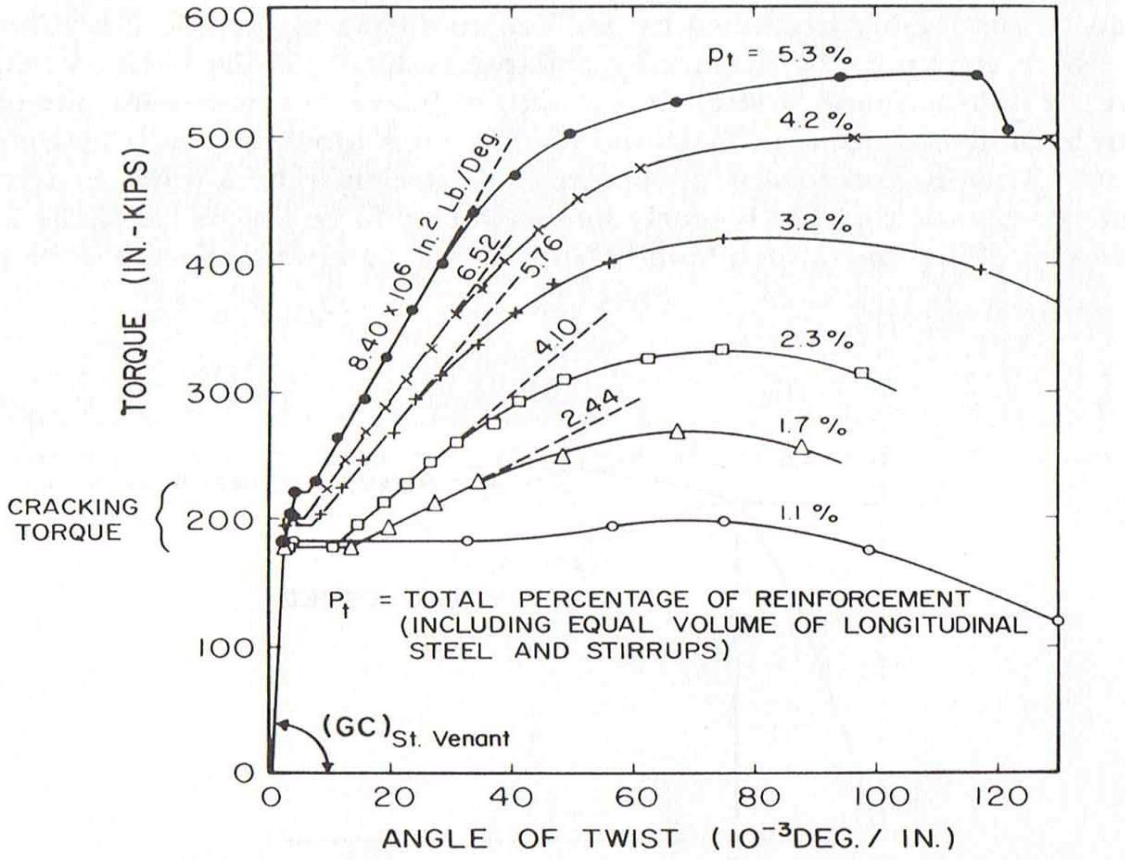
- Where, K_t is section modulus for torsion and it is decided by only shape and size of cross-section

Section Modulus for Torsion

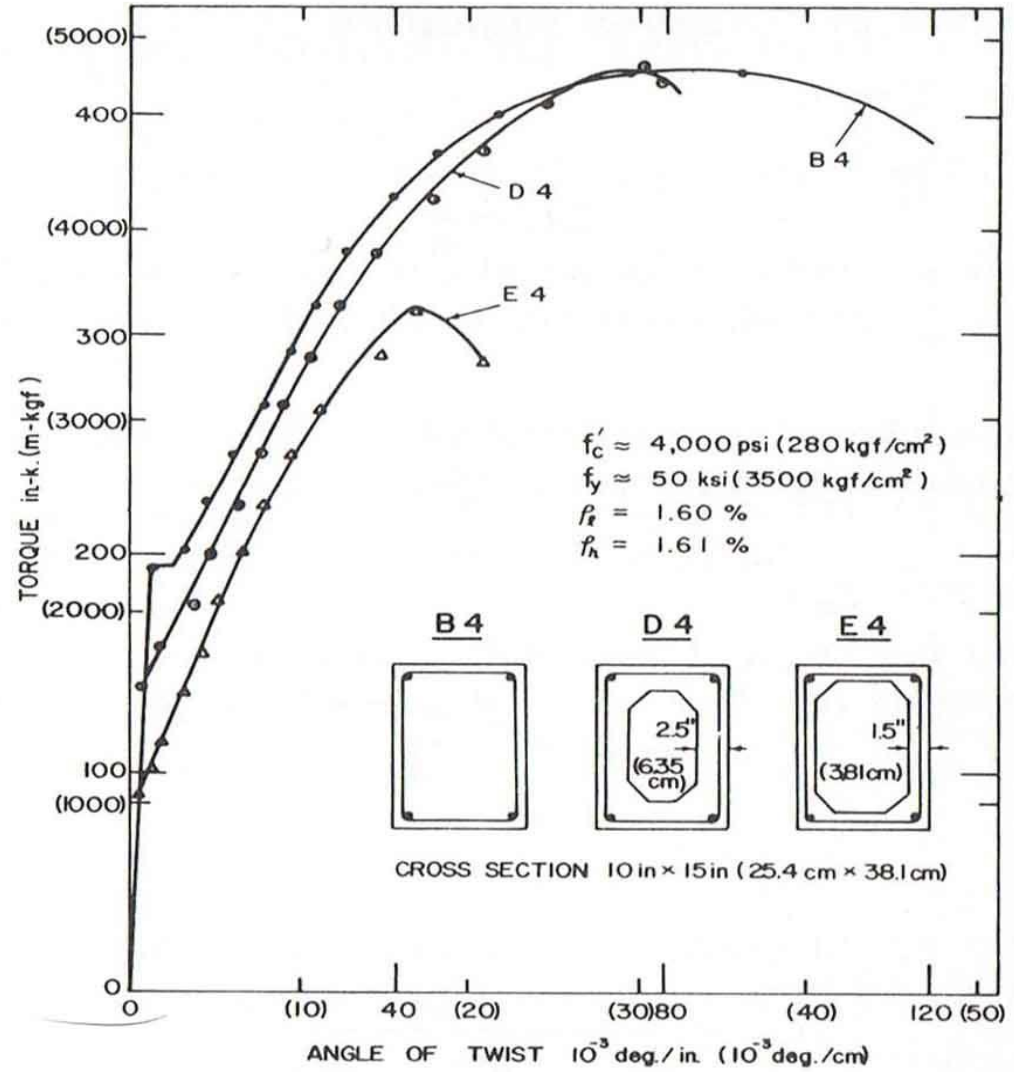
	O:b^2d/n_1 X:$b^2d/(n_1n_2)$	$n_1 = 3.1 + \frac{1.8}{d/b}$ $n2 = 0.7 + \frac{0.3}{d/b}$
	$\sum \frac{b_i^{\,2} d_i}{n_i}$ Bi and di are the length of shorter and longer sides of the components rectangles ,respectively.	It is appropriate to subdivide the cross section into component rectangles in such a way as to result in the highest possible torsional rigidity
	$2A_mt_1$ Kt of a box section shall be calculated as a hollow section. In case the ratio of the member thickness to the entire width of the box section in the direction of thickness exceeds 0.15, Kt may be calculated by taking section as solid case.	Am: area enclosed by the centerline of wall thickness t_1 =web thickness

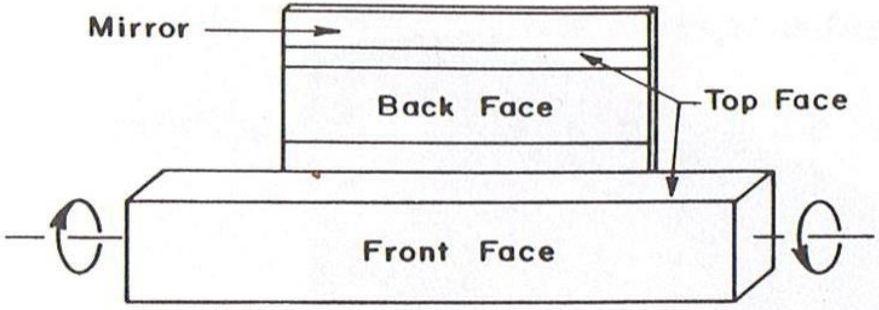
$T - \theta$ Curve (Torque - Twist Behavior)



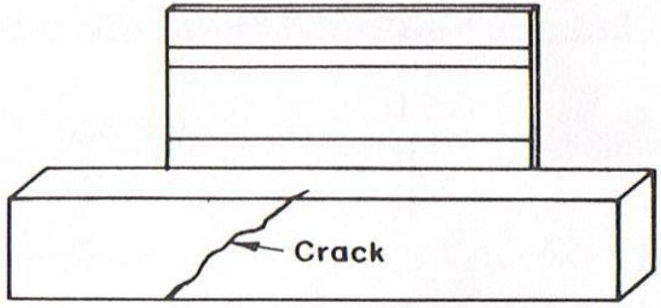


Torque - Twist Behavior

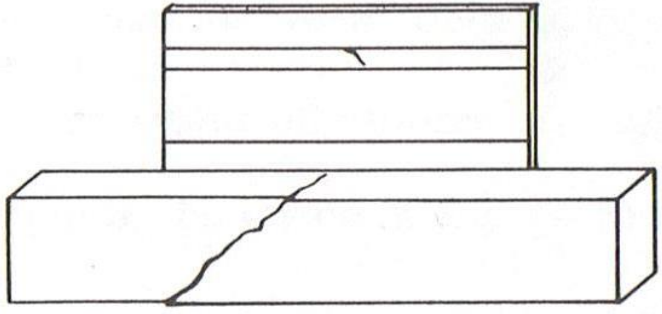




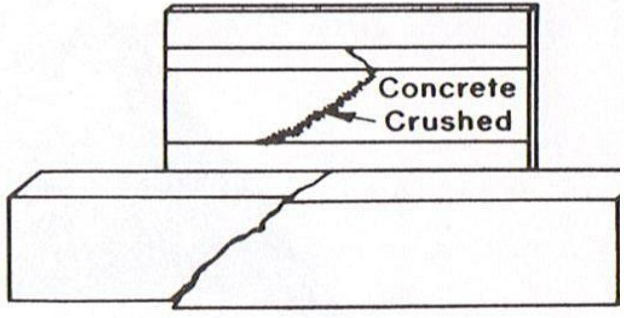
(a) Middle section of beam under torsion. Camera could see front face and top face directly. Back face was reflected in a mirror.



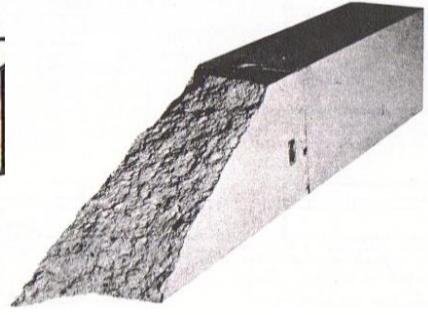
(b) 1/100 second after diagonal crack was picked up by camera. Crack extends across front face.



(c) 1/15 second after the crack was observed. Crack widens and spreads across the top face.

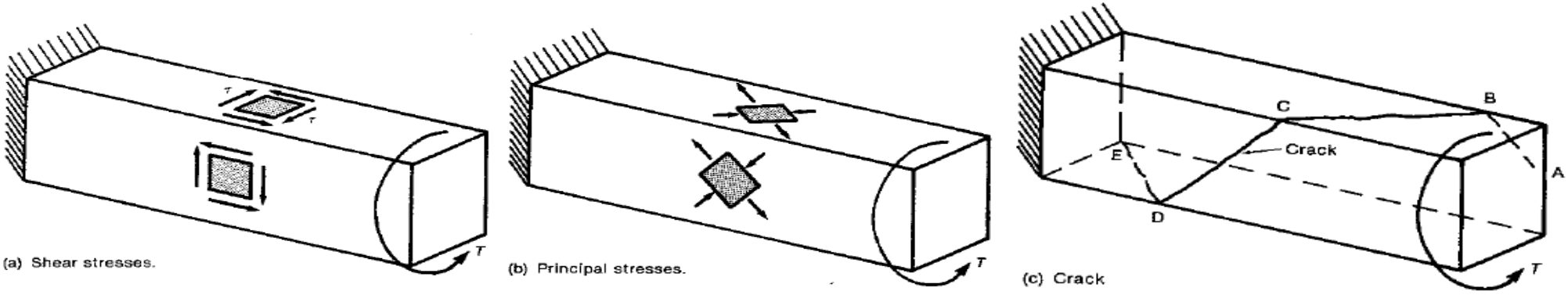


(d) 1/5 second after the crack was observed. Concrete crushes at the back face as shown in the mirror.



Torsional Crack (Unreinforced Concrete)

- Torsional Crack (Unreinforced Concrete)
  - Pure torsional moment act on a member– Pure shear stress



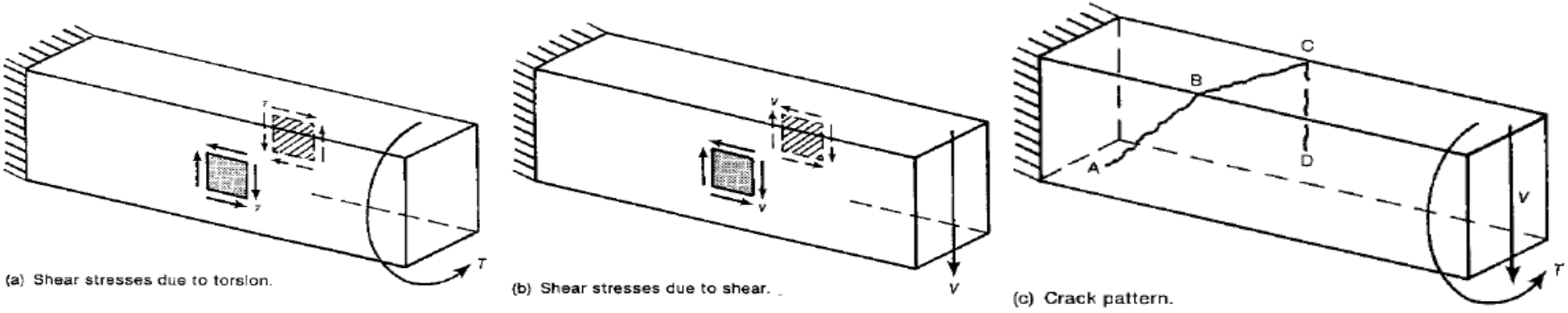
  - From Mohr's stress circle, Principe stresses (σ_1, σ_2)

  - Torsional Crack caused by σ_1 when it exceeds the tensile strength of concrete.

$$\sigma_1 = \sigma_2 = |\tau_{\max}| = \frac{M_t}{K_t} \quad (tension: +)$$

Torsional Crack (Unreinforced Concrete)

- Torsional Crack (Unreinforced Concrete)
  - Combined Torsion and Shear



When a member is subjected to combined shear and torsion, the two shearing stress components add on one side face and counteract each other on other side. As a results, inclined cracking starts on the face where stresses add (Crack AB) and extends across the flexural tensile face of the beam. If the bending moments are sufficiently large, the cracks will extend almost vertically across the back face (Crack CD)

ACI Design for Torsion

Design for Torsion (ACI)

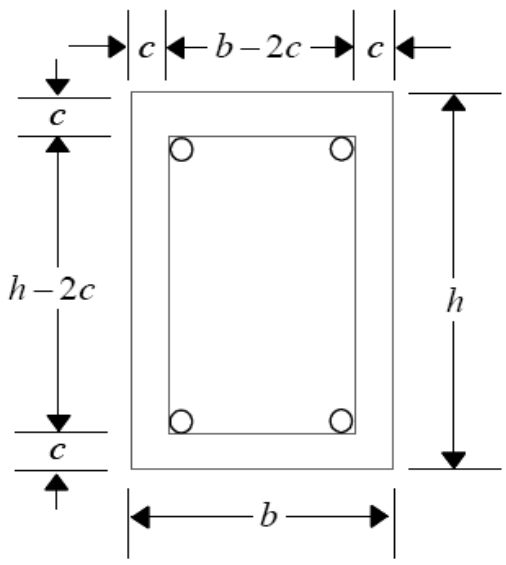
1) Determine the factored torsion, T_u. → From Analysis
2) Determine special section properties.
3) Determine critical torsion capacity.
4) Determine the reinforcement steel required.
5) Check the minimum reinforcement

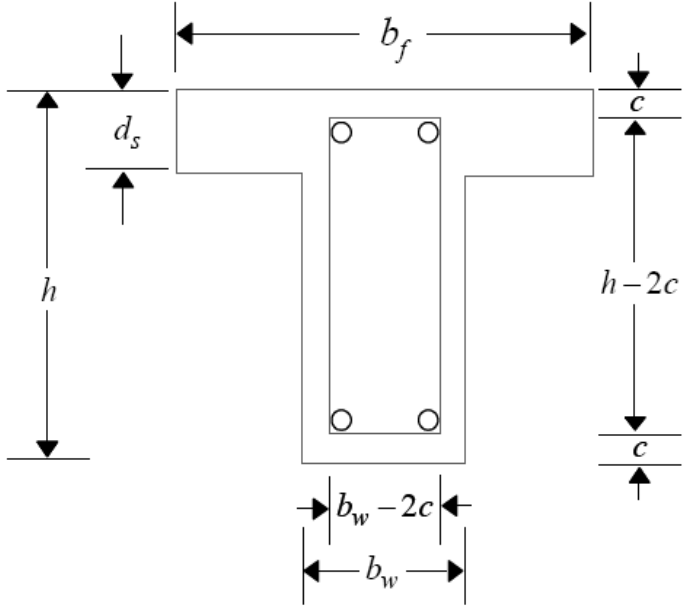
Step 2: Special Section Properties

- A_{cp} = Area enclosed by outside perimeter of concrete cross-section
- A_{oh} = Area enclosed by centerline of the outermost closed transverse torsional reinforcement
- A_o = Gross area enclosed by shear flow path
- p_{cp} = Outside perimeter of concrete cross section
- p_n = Perimeter of centerline of outermost closed transverse torsional reinforcement

Special Section Properties

- For rectangular beam
  - A_{cp} = bh
  - A_{oh} = (b – 2c) (h – 2c)
  - A_o = 0.85 A_{oh}
  - p_{cp} = 2b + 2h
  - p_n = 2 (b – 2c) + 2 (h – 2c)
- For T beam
  - A_{cp} = b_wh + (b_f – b_w) d_s
  - A_{oh} = (b_w – 2c) (h – 2c)
  - A_o = 0.85 A_{oh}
  - p_{cp} = 2b_f + 2h
  - p_n = 2 (b_w – 2c) + 2 (h – 2c)





Closed Stirrup in T-Beam Section

Step 3: Torsion Capacity (T_{cr}) – Requirement of Torsional Steel

22.7.3 *Factored design torsion*

22.7.3.1 If $T_u \geq \phi T_{cr}$ and T_u is required to maintain equilibrium, the member shall be designed to resist T_u.

22.7.3.2 In a statically indeterminate structure where $T_u \geq \phi T_{cr}$ and a reduction of T_u can occur due to redistribution of internal forces after torsional cracking, it shall be permitted to reduce T_u to ϕT_{cr}, where the cracking torsion T_{cr} is calculated in accordance with 22.7.5.

22.7.3.3 If T_u is redistributed in accordance with 22.7.3.2, the factored moments and shears used for design of the adjoining members shall be in equilibrium with the reduced torsion.

Step 3: Torsion Capacity (T_{cr}) – Requirement of Torsional Steel

22.7.5.1 Cracking torsion T_{cr} shall be calculated in accordance with Table 22.7.5.1 for solid and hollow cross sections, where N_u is positive for compression and negative for tension.

Table 22.7.5.1—Cracking torsion

Type of member	T_{cr}	
Nonprestressed member	$4\lambda\sqrt{f'_c}\left(\frac{A_{cp}^2}{p_{cp}}\right)$	(a)
Prestressed member	$4\lambda\sqrt{f'_c}\left(\frac{A_{cp}^2}{p_{cp}}\right)\sqrt{1+\frac{f_{pc}}{4\lambda\sqrt{f'_c}}}$	(b)
Nonprestressed member subjected to axial force	$4\lambda\sqrt{f'_c}\left(\frac{A_{cp}^2}{p_{cp}}\right)\sqrt{1+\frac{N_u}{4A_g\lambda\sqrt{f'_c}}}$	(c)

Step 3: Threshold Torsion Capacity (T_{th}) – [ACI 318-11]

- If the factored torsion is less than threshold torsion, it can be ignored. Threshold torsion is 1/4th of T_{cr}.

(a) For nonprestressed members

$$\phi\lambda\sqrt{f_c'}\left(\frac{A_{cp}^2}{p_{cp}}\right)$$

(b) For prestressed members

$$\phi\lambda\sqrt{f_c'}\left(\frac{A_{cp}^2}{p_{cp}}\right)\sqrt{1+\frac{f_{pc}}{4\lambda\sqrt{f_c'}}}$$

(c) For nonprestressed members subjected to an axial tensile or compressive force

$$\phi\lambda\sqrt{f_c'}\left(\frac{A_{cp}^2}{p_{cp}}\right)\sqrt{1+\frac{N_u}{4A_g\lambda\sqrt{f_c'}}}$$

Step 4: Required Torsional Strength and Reinforcement

- If $T_u > \phi T_{cr}$,

22.7.6.1 For nonprestressed and prestressed members, T_n shall be the lesser of (a) and (b):

$$\text{(a)}\ T_n = \frac{2A_o A_t f_{yt}}{s}\cot\theta \qquad (22.7.6.1a)$$

$$\text{(b)}\ T_n = \frac{2A_o A_\ell f_y}{p_h}\tan\theta \qquad (22.7.6.1b)$$

where A_o shall be determined by analysis, θ shall not be taken less than 30 degrees nor greater than 60 degrees; A_t is the area of one leg of a closed stirrup resisting torsion; A_ℓ is the area of longitudinal torsional reinforcement; and p_h is the perimeter of the centerline of the outermost closed stirrup.

Design Equation

$$\phi T_n > T_u$$

Step 4: Required Torsional Reinforcement

- If $T_u > \phi T_{cr}$,

Design Equation

$$\phi T_n > T_u$$

  - The required longitudinal rebar area is

$$A_l = \frac{T_u p_h}{\phi 2 A_0 f_y \tan\theta}$$

  - The required closed stirrup area is $\frac{A_t}{s} = \frac{T_u \tan\theta}{\phi 2 A_0 f_y s} b_w$

Step 4: Design for Torsional Steel

- Angle is the inclination of compression diagonal in space truss.
  - Its value is between 30 to 60 degree
  - Usually it is 45 degrees

22.7.6.1.2 In Eq. (22.7.6.1a) and (22.7.6.1b), it shall be permitted to take θ equal to (a) or (b):

(a) 45 degrees for nonprestressed members or members with $A_{ps}f_{se} < 0.4(A_{ps}f_{pu} + A_s f_y)$

(b) 37.5 degrees for prestressed members with $A_{ps}f_{se} \geq 0.4(A_{ps}f_{pu} + A_s f_y)$

- For a two-legged stirrup, total area for shear and torsion is

$$\text{Total}\left(\frac{A_{v+t}}{s}\right) = \frac{A_v}{s} + \frac{2A_t}{s}$$

Step 5: Minimum Reinforcement (ACI 318-19)

9.6.4.2 If torsional reinforcement is required, minimum transverse reinforcement $(A_v + 2A_t)_{min}/s$ shall be the greater of (a) and (b):

(a) $0.75\sqrt{f_c'}\dfrac{b_w}{f_{yt}}$

(b) $50\dfrac{b_w}{f_{yt}}$

9.6.4.3 If torsional reinforcement is required, minimum area of longitudinal reinforcement $A_{\ell,min}$ shall be the lesser of (a) and (b):

(a) $$\frac{5\sqrt{f_c'}A_{cp}}{f_y} - \left(\frac{A_t}{s}\right)p_h\frac{f_{yt}}{f_y}$$

(b) $$\frac{5\sqrt{f_c'}A_{cp}}{f_y} - \left(\frac{25b_w}{f_{yt}}\right)p_h\frac{f_{yt}}{f_y}$$

Step 5: Maximum Torsional Moment Strength – [ACI 318-19]

- To reduce unsightly cracking and compressive failure due to shear and torsion, the size of the cross-section is limited.

22.7.7.1 Cross-sectional dimensions shall be selected such that (a) or (b) is satisfied:

(a) For solid sections

$$\sqrt{\left(\frac{V_u}{b_w d}\right)^2 + \left(\frac{T_u p_h}{1.7 A_{oh}^2}\right)^2} \le \phi\left(\frac{V_c}{b_w d} + 8\sqrt{f_c'}\right) \qquad (22.7.7.1a)$$

(b) For hollow sections

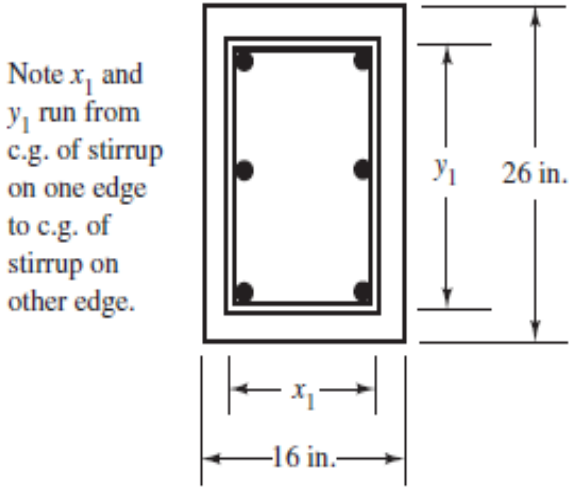
$$\left(\frac{V_u}{b_w d}\right) + \left(\frac{T_u p_h}{1.7 A_{oh}^2}\right) \le \phi\left(\frac{V_c}{b_w d} + 8\sqrt{f_c'}\right) \qquad (22.7.7.1b)$$

Additional ACI Requirements

- Torsion be calculated at a distance "d" from face of support unless required
- Design yield strength of torsional steel should not be more than 60,000 psi.
- The longitudinal steel must be distributed uniformly around inside perimeter of the stirrup. In no case, spacing should exceed 12" and at least one bar be present at corners
- Torsional reinforcement must be provided at a distance $b_t + d$ beyond the point at which torsional reinforcement is no more required. b_t is the width of cross section with torsional steel.

Example: Design for Torsion

Design the torsional reinforcing needed for the beam shown in Figure 15.13 if $f'_c = 4000$ psi, $f_y = 60{,}000$ psi, $T_u = 30$ ft-k, and $V_u = 60$ k. Assume 1.5-in. clear cover, #4 stirrups, and a required A_s for M_u of 3.52 in.^{2}. Select #8 bars for flexural reinforcing. Normal-weight concrete is specified.



Example: Design for Torsion

1. Is Torsion Reinforcing Needed?

A_{cp} = area enclosed by outside perimeter of concrete cross section = (16)(26) = 416 in.^{2}
(the letter *c* stands for concrete, and the letter *p* stands for perimeter of cross section)

p_{cp} = outside perimeter of the cross section
= (2)(16 in. + 26 in.) = 84 in.

Torsion, T_u, can be neglected if less than

$$\phi\lambda\sqrt{f'_c}\frac{A_{cp}^2}{p_{cp}} = (0.75)(1.0)(\sqrt{4000}\text{ psi})\left[\frac{(416\text{ in.}^2)^2}{84\text{ in.}}\right]$$

$$= 97{,}723\text{ in-lb} = 97.72\text{ in-k} < 30\text{ ft-k} \times 12\text{ in/ft} = 360\text{ in-k}$$

∴ Torsion must be considered

Example: Design for Torsion

2. Compute Sectional Properties

A_{oh} = area enclosed by centerline of the outermost closed stirrups

Noting 1.5-in. clear cover and #4 stirrups

$$x_1 = 16 \text{ in.} - (2)\,(1.5 \text{ in.} + 0.25 \text{ in.}) = 12.5 \text{ in.}$$
$$y_1 = 26 \text{ in.} - (2)\,(1.5 \text{ in.} + 0.25 \text{ in.}) = 22.5 \text{ in.}$$

See Figure 15.13

$$A_{oh} = (12.5 \text{ in.})\,(22.5 \text{ in.}) = 281.25 \text{ in.}^2$$

A_o = gross area enclosed by shear flow path

$$= 0.85A_{oh} \text{ (from ACI Section 11.5.3.6)}$$
$$= (0.85)\,(281.25 \text{ in.}^2) = 239.06 \text{ in.}^2$$

d = effective depth of beam

$$= 26 \text{ in.} - 1.50 \text{ in.} - 0.50 \text{ in.} - \frac{1.00 \text{ in.}}{2} = 23.50 \text{ in.}$$

p_h = perimeter of centerline of outermost closed torsional reinforcing

$$= (2)\,(x_1 + y_1)$$
$$= (2)\,(12.5 \text{ in.} + 22.5 \text{ in.}) = 70 \text{ in.}$$

Example: Design for Torsion

3. Is the Concrete Section Sufficiently Large to Support the Torsion?

V_c = nominal shear strength of concrete section

$$= 2\lambda\sqrt{f'_c}b_w d = (2)(1.0)(\sqrt{4000}\text{ psi})(16\text{ in.})(23.50\text{ in.}) = 47{,}561\text{ lb}$$

Applying ACI Equation 11-18

$$\sqrt{\left(\frac{V_u}{b_w d}\right)^2 + \left(\frac{T_u p_h}{1.7A_{oh}^2}\right)^2} \le \phi\left(\frac{V_c}{b_w d} + 8\sqrt{f'_c}\right)$$

$$\sqrt{\left(\frac{60{,}000\text{ lb}}{16\text{ in.} \times 23.50\text{ in.}}\right)^2 + \left[\frac{360{,}000\text{ in-lb} \times 70\text{ in.}}{1.7 \times (281.25\text{ in.}^2)^2}\right]^2} \le 0.75\left(\frac{47{,}561\text{ lb}}{16\text{ in.} \times 23.50\text{ in.}} + 8\sqrt{4000}\text{ psi}\right)$$

$$246\text{ psi} < 474\text{ psi}$$

$\therefore$ Section is sufficiently large

4. Determine the Transverse Torsional Reinforcing Required

$$T_n = \frac{T_u}{\phi} = \frac{30\text{ ft-k}}{0.75} = 40\text{ ft-k} = 480{,}000\text{ in-lb}$$

Assuming $\theta = 45^\circ$ as per ACI Section 11.5.3.6(a)

$$\frac{A_t}{s} = \frac{T_n}{2A_o f_{yt}\cot\theta} = \frac{480{,}000\text{ in-lb}}{(2)(239.06\text{ in.}^2)(60{,}000\text{ psi})(\cot 45^\circ)}$$

$$= 0.0167\text{ in.}^2/\text{in. for one leg of stirrups}$$

Example: Design for Torsion

5. Calculate the Area of Shear Reinforcing Required

$$V_u = 60{,}000 \text{ lb} > \frac{1}{2}\phi V_c = \left(\frac{1}{2}\right)(0.75)(47{,}561 \text{ lb}) = 17{,}835 \text{ lb}$$

$\therefore$ Shear reinforcing is required

$$V_s = \frac{V_u - \phi V_c}{\phi} = \frac{60{,}000 \text{ lb} - (0.75)(1.0)(47{,}561 \text{ lb})}{0.75} = 32{,}439 \text{ lb}$$

Applying ACI Equation 11-15

$$\frac{A_v}{s} = \frac{V_s}{f_{yt} d} = \frac{32{,}439 \text{ lb}}{(60{,}000 \text{ psi})(23.50 \text{ in.})}$$

$$= 0.0230 \text{ in.}^2/\text{in. for 2 legs of stirrup}$$

6. Select Stirrups

Total web reinforcing required for two legs $= \frac{A_{v+t}}{s}$

$$= \frac{2A_t}{s} + \frac{A_v}{s} = (2)(0.0167 \text{ in.}^2/\text{in.}) + 0.0230 \text{ in.}^2/\text{in.}$$

$$= 0.0564 \text{ in.}^2/\text{in. for two legs}$$

Using #4 stirrups

$$s = \frac{(2)(0.20 \text{ in.}^2)}{0.0564 \text{ in.}^2/\text{in.}} = 7.09 \text{ in. o.c.} \qquad \text{Say 7 in.}$$

Maximum allowable spacing of stirrups

$$\text{from ACI Section 11.5.6.1} = \frac{p_h}{8} = \frac{70 \text{ in.}}{8} = \text{Say 8.75 in. or 12 in.}$$

Example: Design for Torsion

Minimum area of stirrups A_v by ACI Equation 11-23

$$A_v + 2A_t = 0.75\sqrt{f'_c}\frac{b_w s}{f_{yt}} < \frac{50 b_w s}{f_{yt}} = \frac{(50)(16\text{ in.})(7\text{ in.})}{60{,}000\text{ psi}} = 0.0933\text{ in.}$$

$$= 0.75\sqrt{4000}\frac{(16\text{ in.})(7\text{ in.})}{60{,}000\text{ psi}} = 0.089\text{ in.}^2$$

$$< (2)(0.20\text{ in.}^2) = 0.40\text{ in.}^2 \quad \underline{\text{OK}}$$

Use #4 stirrups @ 7 in.

7. Selection of Longitudinal Torsion Reinforcing

Additional longitudinal reinforcing required for torsion

$$A_\ell = \frac{A_t}{s} p_h \left(\frac{f_{yt}}{f_y}\right)\cot^2\theta \qquad \text{(ACI Equation 11-22)}$$

$$= (0.0167\text{ in.}^2/\text{in.})(70\text{ in.})\left(\frac{60{,}000\text{ psi}}{60{,}000\text{ psi}}\right)(1.00)^2 = 1.17\text{ in.}^2$$

$$\text{Min. } A_\ell = \frac{5\sqrt{f'_c}A_{cp}}{f_y} - \left(\frac{A_t}{s}\right)p_h\frac{f_{yt}}{f_y} \qquad \text{(ACI Equation 11-24)}$$

$$\left[\text{with min } \frac{A_t}{s} = \frac{25 b_w}{f_{yv}} = \frac{(25)(16\text{ in.})}{60{,}000\text{ psi}} = 0.00667\text{ in.}^2/\text{in.}\right]$$

$$= \frac{(5\sqrt{4000}\text{ psi})(416\text{ in.}^2)}{60{,}000\text{ psi}} - (0.0167\text{ in.}^2/\text{in.})(70\text{ in.})\left(\frac{60{,}000\text{ psi}}{60{,}000\text{ psi}}\right)$$

$$= 1.02\text{ in.}^2 < 1.17\text{ in.}^2 \quad \underline{\text{OK}}$$

Example: Design for Torsion

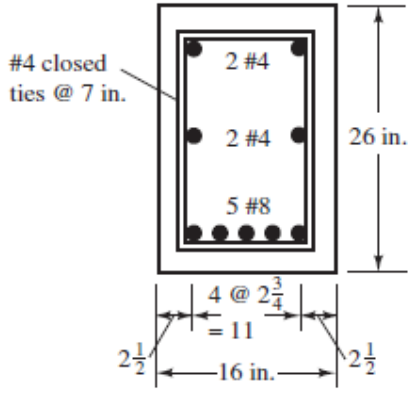


FIGURE 15.14 Reinforcing selected for beam of Example 15.1.

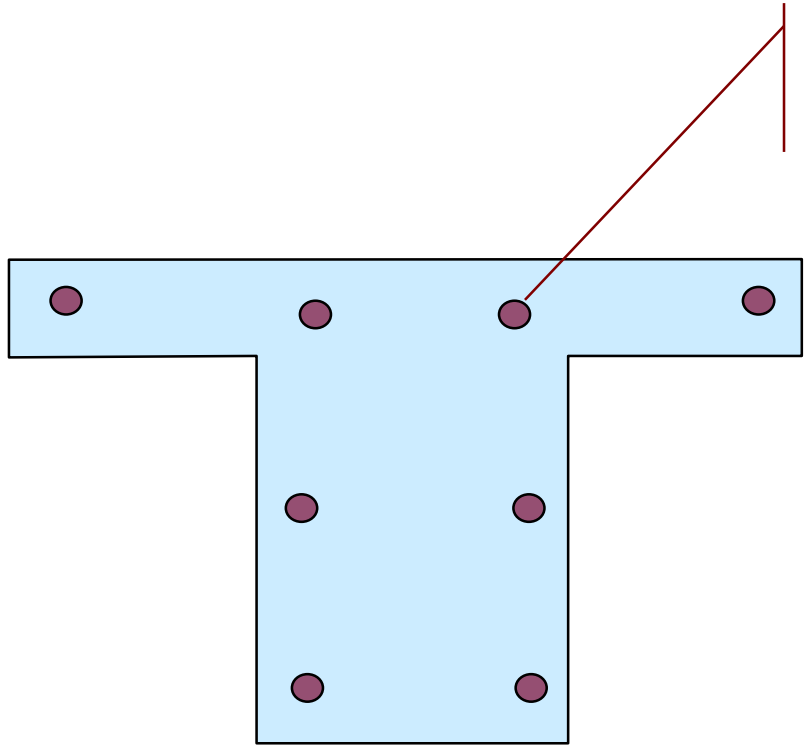
This additional longitudinal steel is spread out into three layers to the four inside corners of the stirrups and vertically in between, with a spacing not in excess of 12 in.

Assume one-third in top = 1.17 in.^{2}/3 = 0.39 in.^{2}, 0.39 in.^{2} + 3.52 in.^{2} = 3.91 in.^{2} in bottom, and remainder 0.39 in.^{2} in between.

Use two #4 bars (0.40 in.^{2}) in top corner, two #4 bars in the middle, and five #8 bars (3.93 in.^{2}) at the bottom, as shown in Figure 15.14.

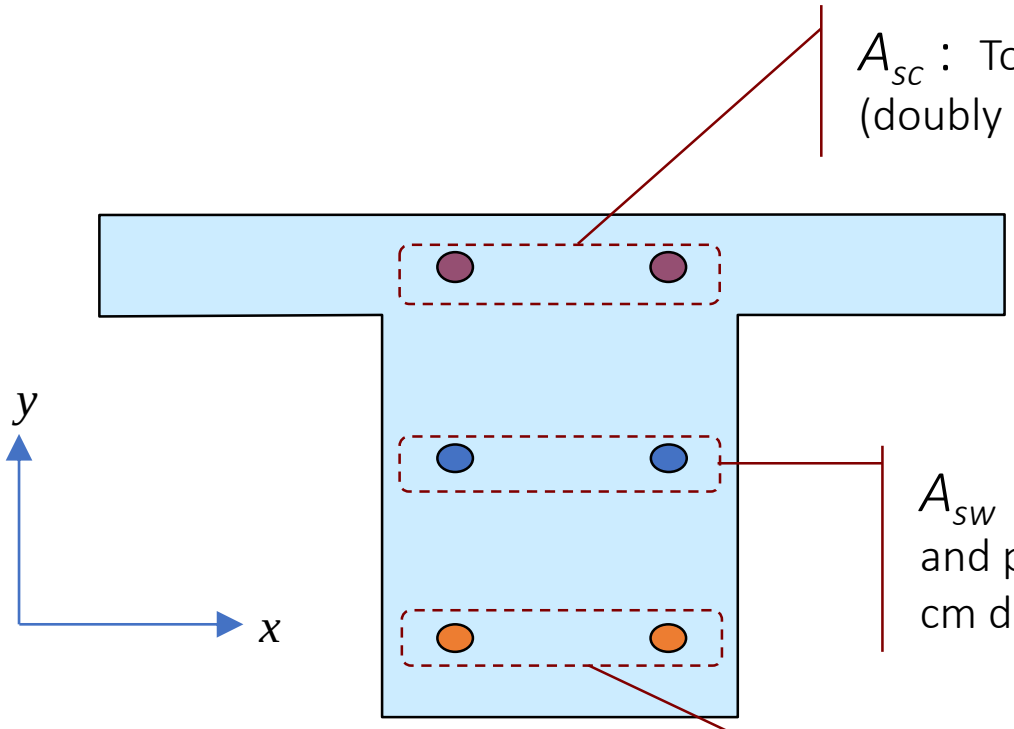
Rebars for Axial, Flexure and Shear Actions

Rebars for Axial Load - P



A_s : May be needed to resist compression.
Generally not required if P is small

Rebars for Moment - M_x

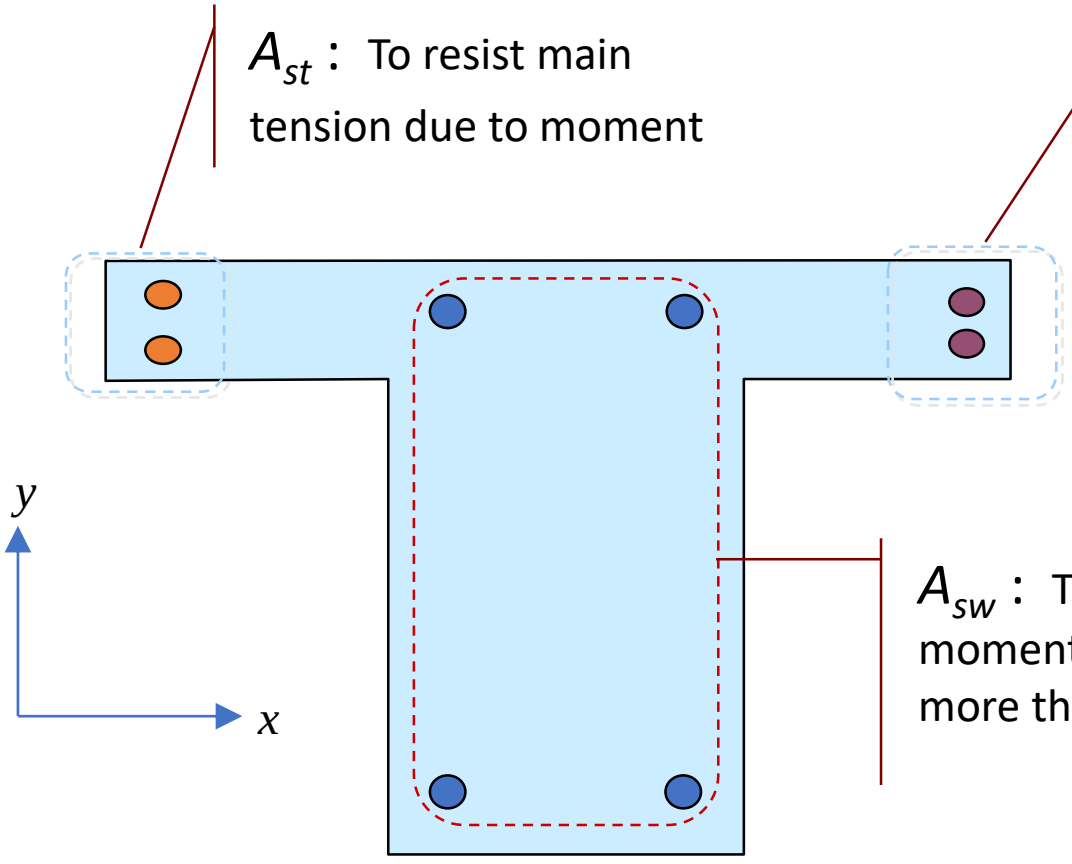


A_{sc} : To resist compression due to moment (doubly reinforced beams)

A_{sw} : To resist secondary tension due to moment and prevent web cracks (for beams more than 90 cm deep)

A_{st} : To resist main tension due to moment

Rebars for Moment - M_y

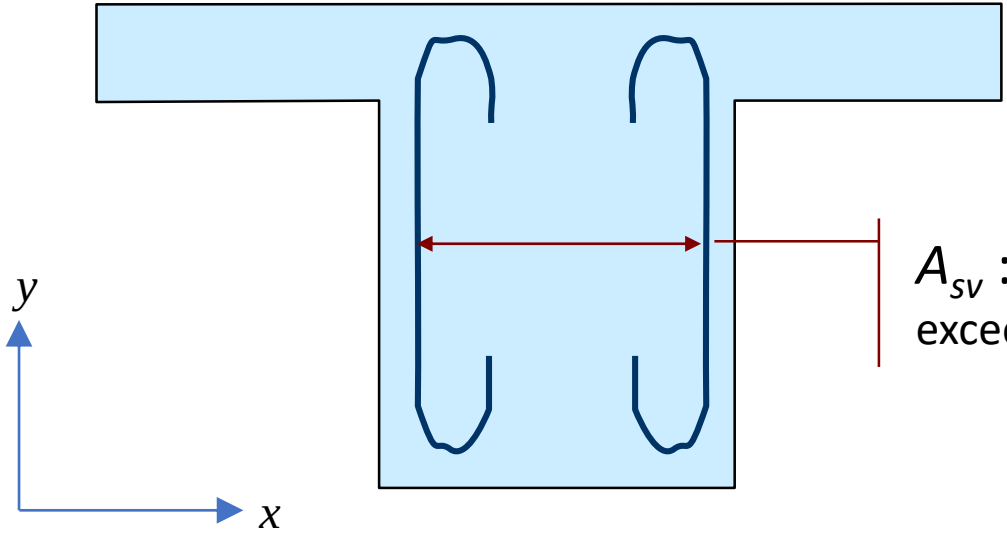


A_{st} : To resist main tension due to moment

A_{sc} : To resist compression due to moment (may not be neede)

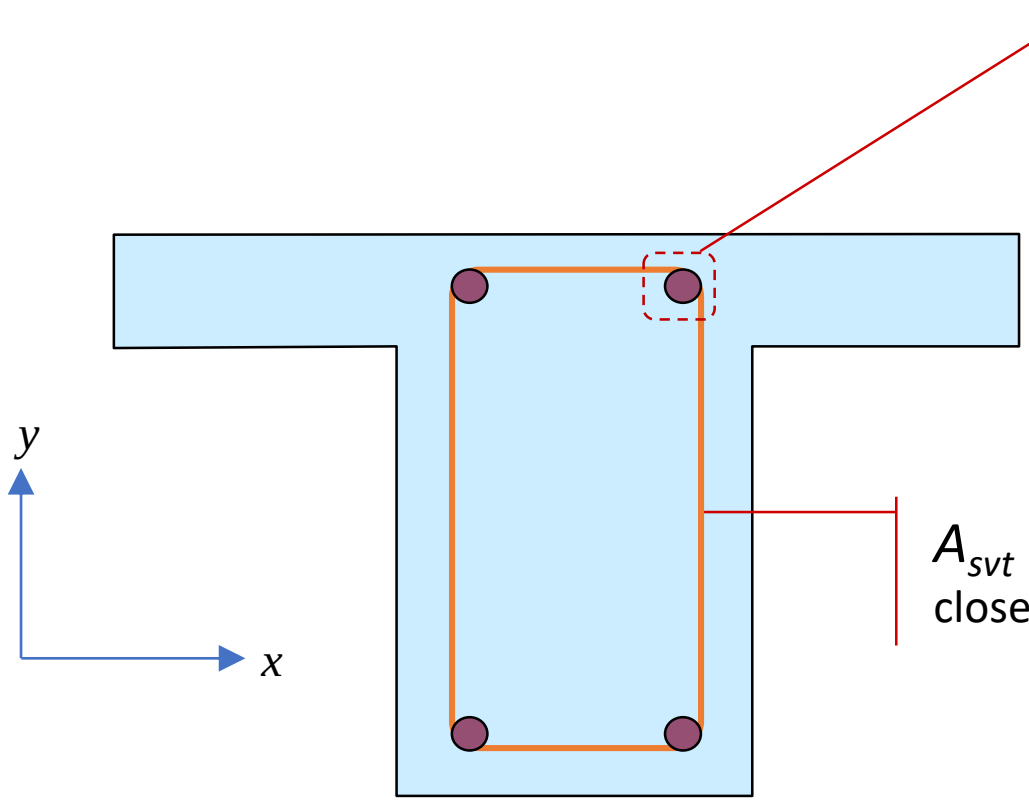
A_{sw} : To resist secondary tension due to moment and prevent web cracks (for beams more than 90 cm deep)

Rebars for Shear - V



A_{sv} : To resist shear stress due to shear force exceeding the shear capacity of concrete

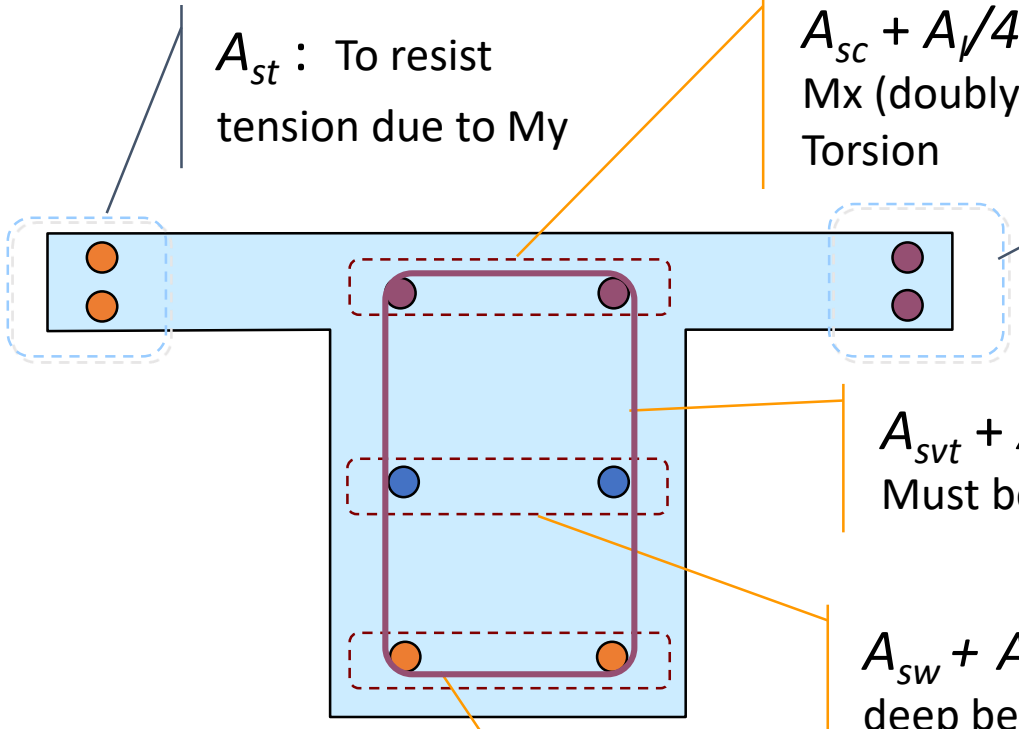
Rebars for Torsion - T



A_l : To resist the longitudinal tension due to Torsion. Must be distributed around the perimeter

A_{svt} : To resist shear due to Torsion. Must be closed hoops on sides of the section

Rebars for P + Mx + My + V + T



A_{st} : To resist tension due to My

$A_{sc} + A_l/4$: To resist compression due to moment Mx (doubly reinforced beams) and tension due to Torsion

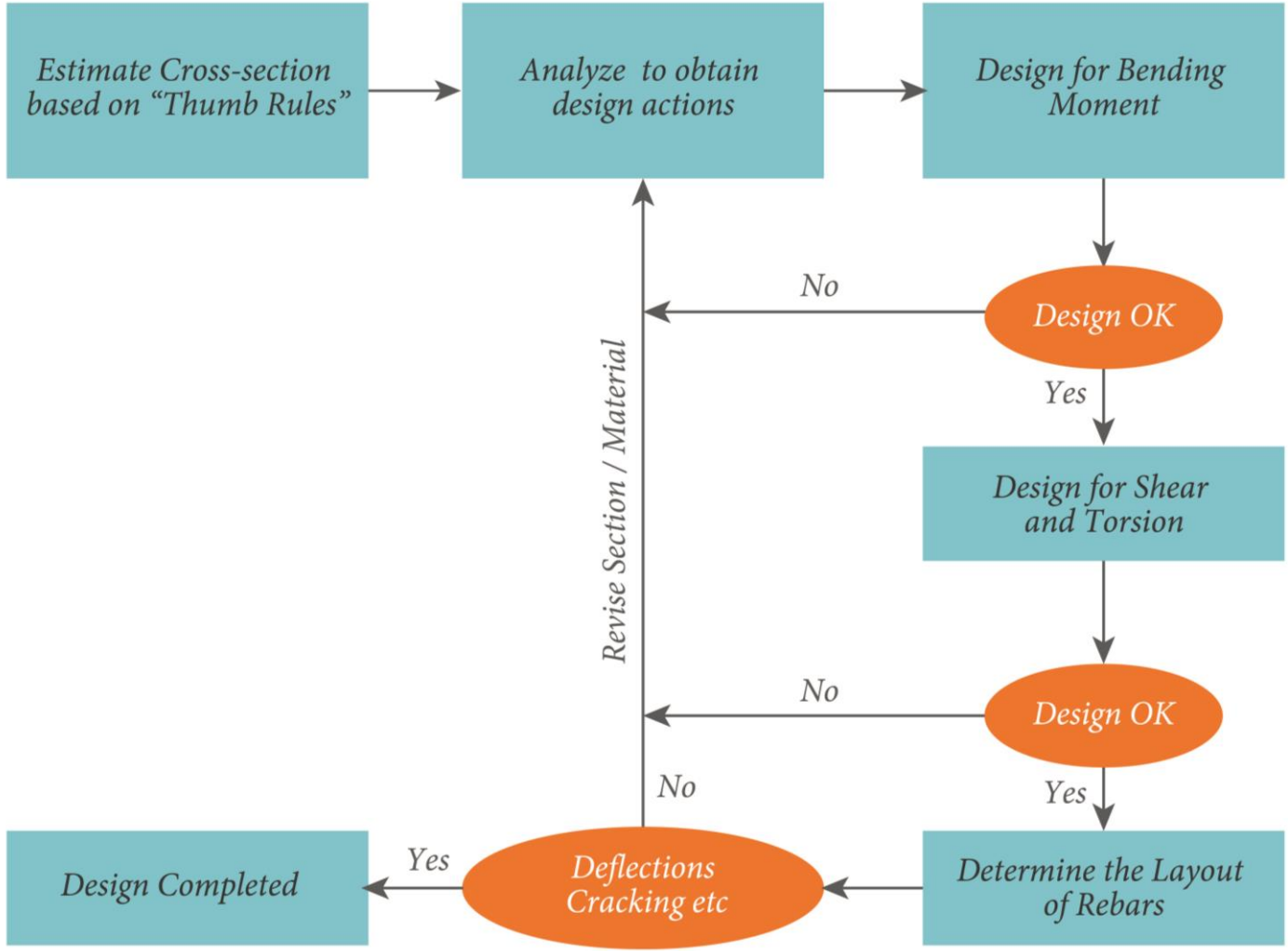
A_{sc} : To resist compression due to My (may not be needed)

$A_{svt} + A_v/2$: To resist shear due to Torsion. Must be closed hoops on sides of the section

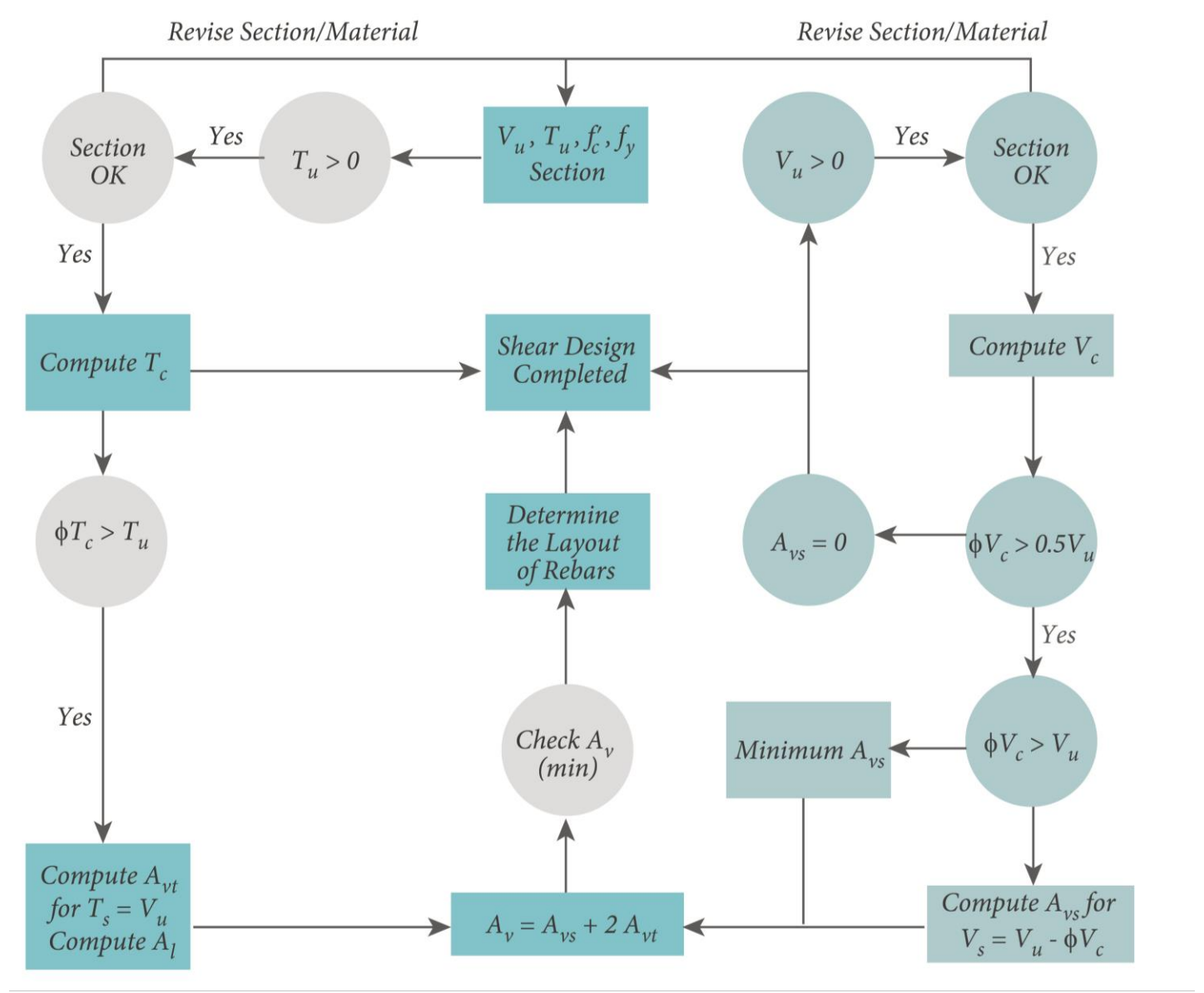
$A_{sw} + A_l/4$: To resist secondary tension in deep beams due to moment and due to Torsion

$A_{st} + A_l/4$: To resist main tension due to moment and tension due to Torsion

Design of Beams: Basic Procedure



Shear-Torsion Design Procedure - ACI




Thank You!